Data Science

Lecture 02: Probability Theory

Prof. Dr. Michael Granitzer

Speaker notes

Converted from LNKB-tex part-en-ds-chp2-probability-and-statistics.tex. The content units of the original chapter are included below in the original order, as shared units from content/units/math/stochastic/ (the same units feed the probabilistic machine learning course), plus the Bernoulli/binomial split-off, sampling and estimation before the test, and course-local parts (module takeaways, the churn shop).

Two sessions. Session 1 covers Modules 0–4 (orientation, events, counting, the axioms, conditioning and Bayes); session 2 covers Modules 5–7 (random variables, describing a sample, testing). The Module 5 slide marks the boundary.

Where We Left Off

Lecture 01 ended on a claim: data does not speak for itself — an assumption always speaks for it, and a pattern is knowledge only once it has been checked. Today builds the language in which both are written.

Assume

Modules 1–5 — a model of chance: outcomes, probabilities, random variables, the joint

→

Observe

Module 6 — a sample, described, with its error

→

Check

Module 7 — would chance alone produce this?

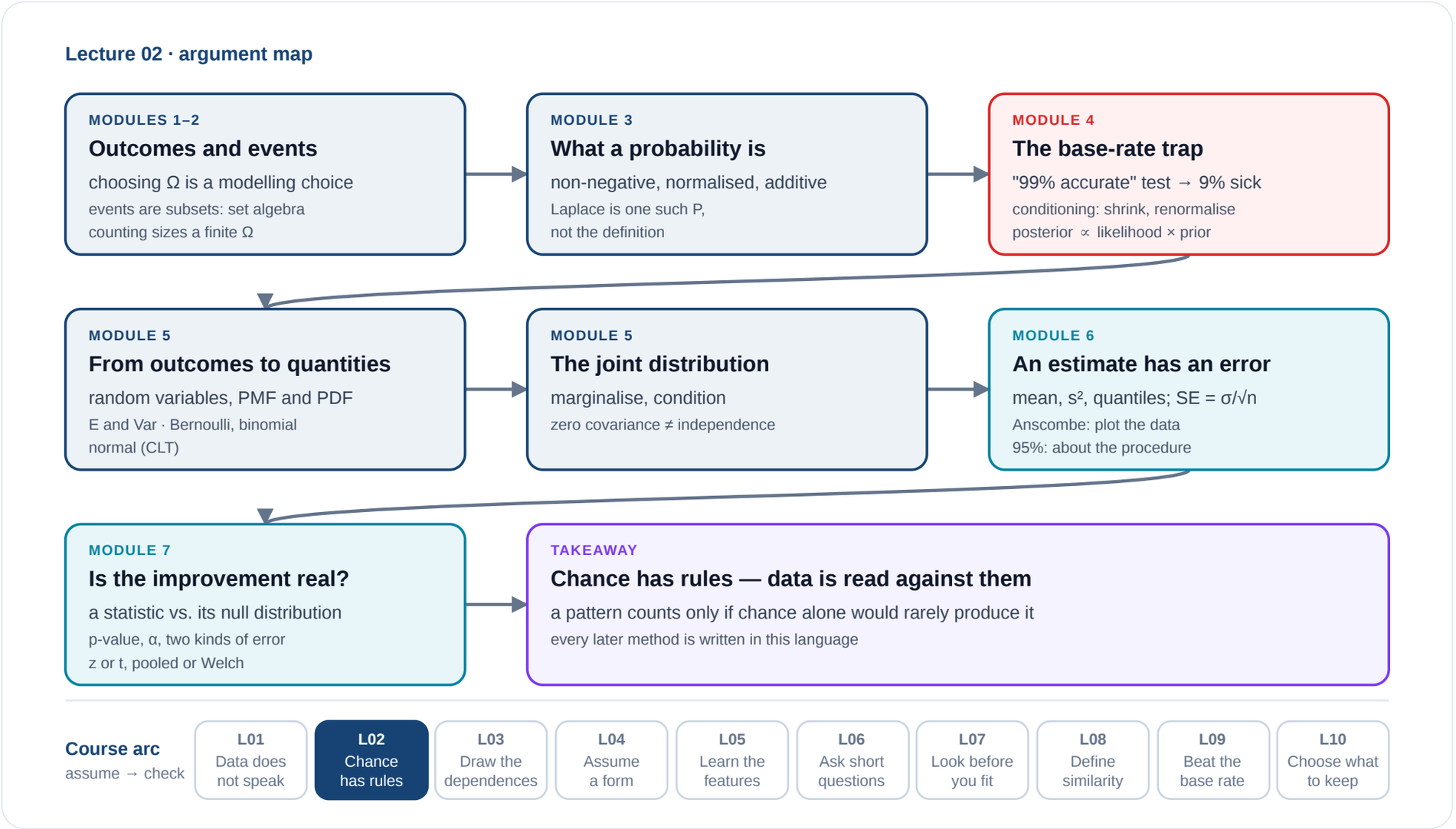
> The app users of Lecture 01 churn less. By the end of today you can say whether that difference is real — not yet *why* (that is Lecture 03).

Speaker notes

Pick up the thread from the roadmap table of Lecture 01. Say which module equips which step. Say why the formal treatment comes first: every later check — a test, a validation error, lift — is a probability statement, so the language has to exist before the first check can be written down.

The lecture spans two sessions; the Module 5 slide marks the boundary. Recall this three-card flow there.

Argument Map



Speaker notes

The lecture on one page. Top row: the objects (outcomes, events), what makes a number a probability, and the one operation that matters, conditioning — with the trap it resolves: the 99% test that is right only 9% of the time. Middle row: from outcomes to quantities to the joint, then to a sample and the error of what we compute from it. Bottom row: the check, a test statistic against its distribution under "chance alone". Point at the strip: every later lecture writes its assumption and its check in this language.

Module 0

Orientation

> ***Guiding Question:*** *What does a data scientist actually need from probability theory — and why does the formal treatment come first?*

The Landscape

Probability theory
Given a model of chance, what follows?
Kolmogorov axioms, distributions, limit theorems

Mathematical statistics
Given data, which model of chance is plausible?
Estimation, hypothesis testing

Data science
Given data and a decision, what should we do?
Prediction, learning, evaluation

The arrow runs left to right: this lecture builds the vocabulary the rest of the course spends its time using.

Speaker notes

Orientation slide. The three columns are not three fields to memorise but three questions — deduction from a model, inference of a model, decision with a model. Say explicitly that we start on the left because everything on the right is stated in its language.

What We Need From Probability

In this lecture

- sample spaces and events
- counting outcomes
- probability as a measure
- conditioning and independence
- Bayes' theorem
- random variables and distributions
- describing data numerically
- testing a hypothesis

Where it comes back

- naive Bayes and Bayesian decisions
- impurity measures in decision trees
- likelihoods in probabilistic models
- sampling and error estimates in evaluation
- independence assumptions in graphical models



Speaker notes

Set expectations: this is the one heavily formal lecture, and every item on the left has a named consumer on the right. If students ask "why do we need this", point at the right-hand column rather than defending rigour in the abstract.

Example: Two Fruit Boxes

Red box
2 apples, 6 oranges

Blue box
3 apples, 1 orange

We pick a box, then a fruit from it.

- If we pick the **red box 40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

Speaker notes

Deliberate overview slide (P4): the two questions are the whole chapter in miniature. Do not formalise anything yet — the point is that both questions are answerable by counting, and that the second one feels harder than the first.

The Counts, in One Grid

A hundred picks with the red box chosen 40% of the time, as a table of counts — box across, fruit down:

	$B = r$	$B = b$	row total
$F = a$ (apple)	10	45	**55**
$F = o$ (orange)	30	15	**45**
column total	**40**	**60**	**100**

Joint
a single cell over the total
$P(B = r, F = a) = \frac{10}{100}$

Marginal
a row or column total over the total
$P(F = a) = \frac{55}{100}$

Conditional
a cell over its own row or column
$P(F = a \mid B = r) = \frac{10}{40}$

Speaker notes

The counts are what the boxes' contents would produce on average: of 40 red picks a quarter are apples, of 60 blue picks three quarters. Everything on this slide is division of counts. Have students compute one of each kind before naming them. The names will be redefined formally later; here they are labels for three ways of dividing the same table.

The Two Rules

Sum rule

$$P(X) = \sum_{Y} P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y \mid X) P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

Speaker notes

This is the Bishop framing: two rules, everything else derived. Students who carry only these two out of the chapter are in good shape — the formal apparatus that follows exists to justify them and to extend them beyond finite tables.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

Which box did the orange come from?

Speaker notes

Bishop's numbers: the answer moves from 0.4 to 2/3, and the table on the previous slide confirms it (30 of 45 oranges came from red). Then the follow-up question: with 3 apples and 9 oranges in blue, both boxes are 75% oranges and the posterior is the prior, 0.4 — the fruit says nothing about the box. That is the deeper lesson: evidence updates belief only to the extent that the hypotheses predict different things.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{\frac{6}{8} \cdot 0.4 + \frac{1}{4} \cdot 0.6} = \frac{2}{3}$$

Before the pick: 40% red. After seeing an orange: 67% red. The orange is **evidence** — because the red box is full of oranges and the blue one is not.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

Which box did the orange come from?

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Before the pick: 40% red. After seeing an orange: 67% red. The orange is **evidence** — because the red box is full of oranges and the blue one is not.

And if the blue box held 3 apples and 9 oranges — the same orange share as the red one?

Takeaway

> Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. What follows makes them precise and extends them beyond counting.

Speaker notes

Signpost explicitly: everything just used informally gets a definition in the coming modules. This unit is the map, not the territory.

Module 1

Events and the Algebra of Sets

> ***Guiding Question:*** *If probability is going to be a measure, what exactly are the objects it measures — and how do they combine?*

What Makes an Experiment "Random"?

Definition (Random experiment). A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Speaker notes

Note what the definition does *not* say: nothing about the world being indeterministic. Ask students for a "natural random experiment" — rainfall, click-through, a queue — where nobody designed the configuration.

Randomness as Missing Information

> A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.
>
> Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Speaker notes

This is the epistemic-modeling principle in probability form (P3): calling something random is a statement about our knowledge, not only about the system. Good place to ask where that matters in data science — noise vs. unmodelled signal is the same distinction.

Sample Space and Events

Definition (Sample space, event space). A set $\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$. The elements are called **outcomes**.

Let Ω be finite. Every subset $A \subseteq \Omega$ is an **event**; the event A **occurs** iff the observed outcome ω is a member of A. The set of all events, the power set $\mathcal{P}(\Omega)$, is the **event space**.

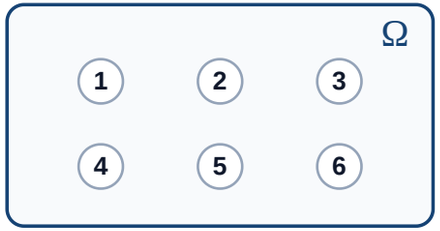
An outcome is a single result; an event is a *question you can ask* about the result.

For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set; we return to this when continuous distributions appear.

Speaker notes

Math first (P5). The load-bearing word is "exactly one" — outcomes must be mutually exclusive and exhaustive, which is a modelling decision, not a mathematical one. The outcome/event distinction is the one students blur.

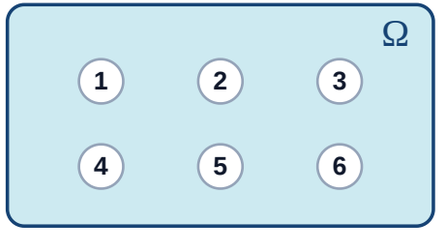
Important Event Types



$\varnothing$

impossible event

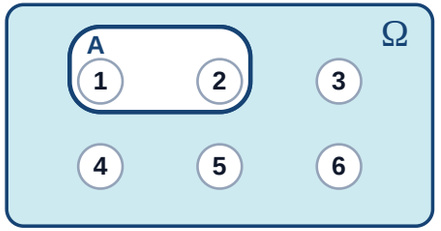
no outcome belongs to it



Ω

certain event

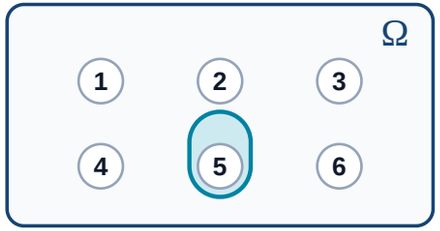
every outcome belongs to it



$\bar{A} = \Omega \setminus A$

complementary event

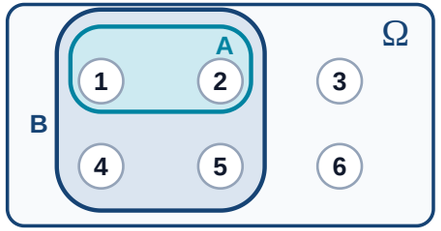
A = {1, 2}, $\bar{A}$ = {3, 4, 5, 6}



$|A| = 1$

elementary event

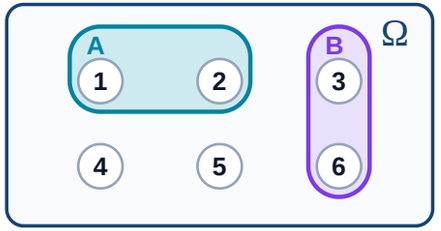
A = {5}: one outcome



$A \subseteq B$

A entails B

A = {1, 2}, B = {1, 2, 4, 5}



$A \cap B = \varnothing$

incompatible events

A = {1, 2}, B = {3, 6}

Speaker notes

Six panels on one sample space, the six faces of a die, so that every event is a set of visible outcomes; read them as vocabulary, not results. "Entails" is worth dwelling on: set inclusion is logical implication between events, which is why the next unit can do algebra with pictures.

Examples

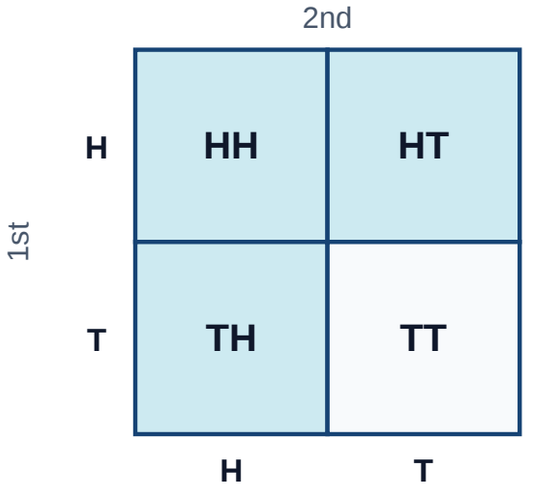
One coin flip



E = {H}: "heads"

all events: ∅, {H}, {T}, {H, T}

Two coin flips



"at least one head"

= {HH, HT, TH}

4 outcomes

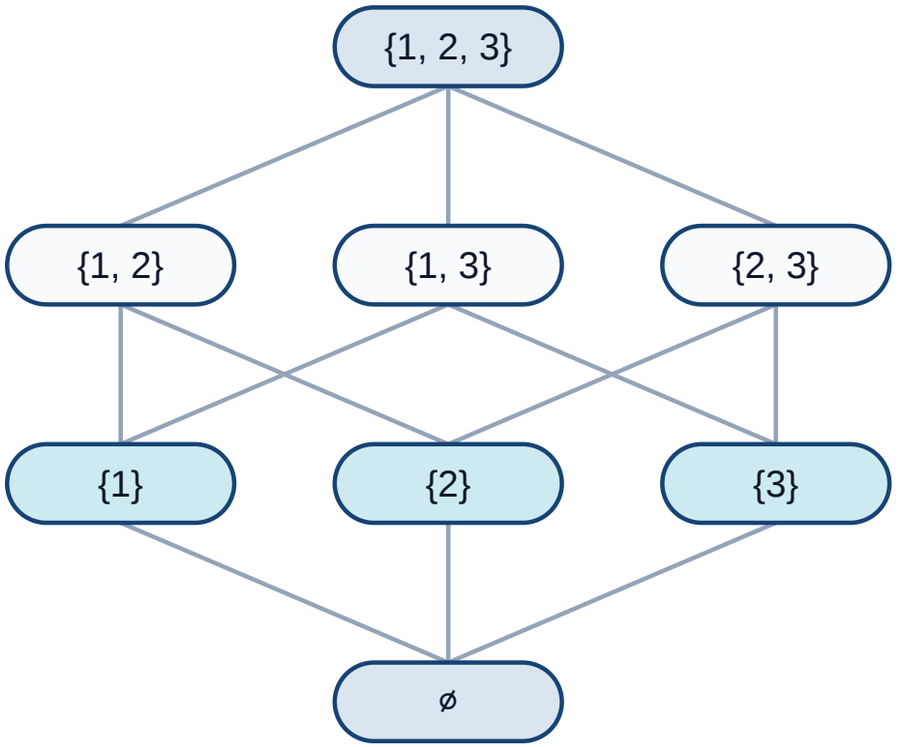
2^4 = 16 possible events

An event is a set of outcomes you can shade — "heads", "at least one head" — and the sample space fixes which such questions can be asked.

Speaker notes

Have students write down the event "exactly one head" ({HT, TH}) and the event "the first flip is heads" ({HH, HT}) — same sample space, different questions.

The Event Space Grows Fast



$|\Omega| = n \rightarrow 2^n$ events

1	2
2	4
3	8
6	64
10	1,024
20	1,048,576

lines: one element added

For $|\Omega| = n$ the event space has 2^n elements — far too many to list, which is why probability is done by counting, not by enumeration.

Speaker notes

The lattice diagram of the original is replaced by three inclusion levels; the point is the 2^n growth, not the drawing. This motivates counting and, later, why we cannot always take the full power set for infinite Ω.

Not Every Sample Space Is Finite

Dosage of a drug: Ω = (0, ∞)



uncountably many outcomes — probability can only live on intervals, not on points

Race with 7 teams: all orderings



7! = 5,040 outcomes; shaded: the event "team 3 wins"

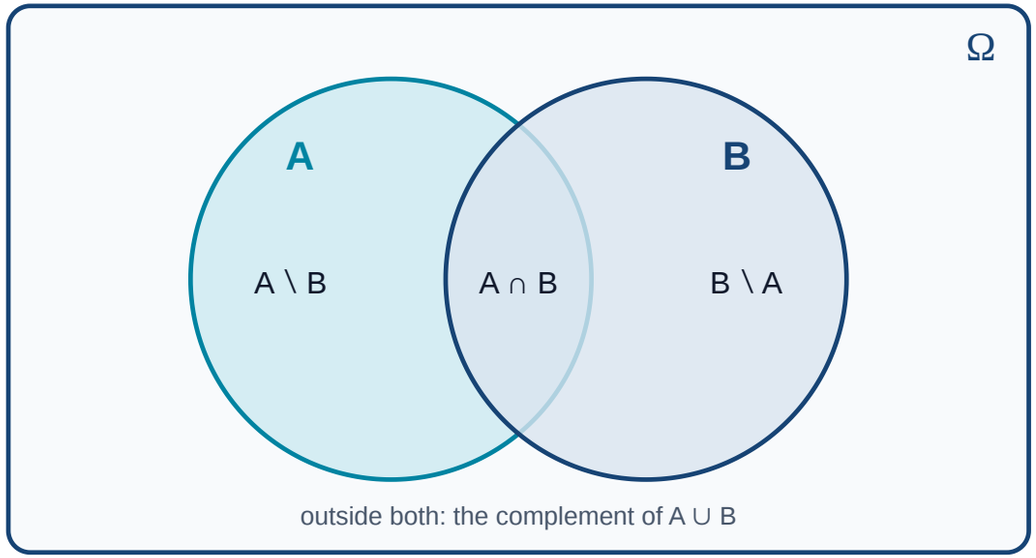
Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Speaker notes

Close the module on the modelling move (P3). Ask what is lost by modelling dosage as a positive real (measurement precision, dosing intervals) — the same question returns for every feature in every dataset later in the course.

Venn Diagrams

Venn diagrams are the graphical representation of events:



- the **rectangle** is the sample space Ω
- a **circle** is an event $A \subseteq \Omega$
- every **region** is a set — and every set operation is a region you can point at

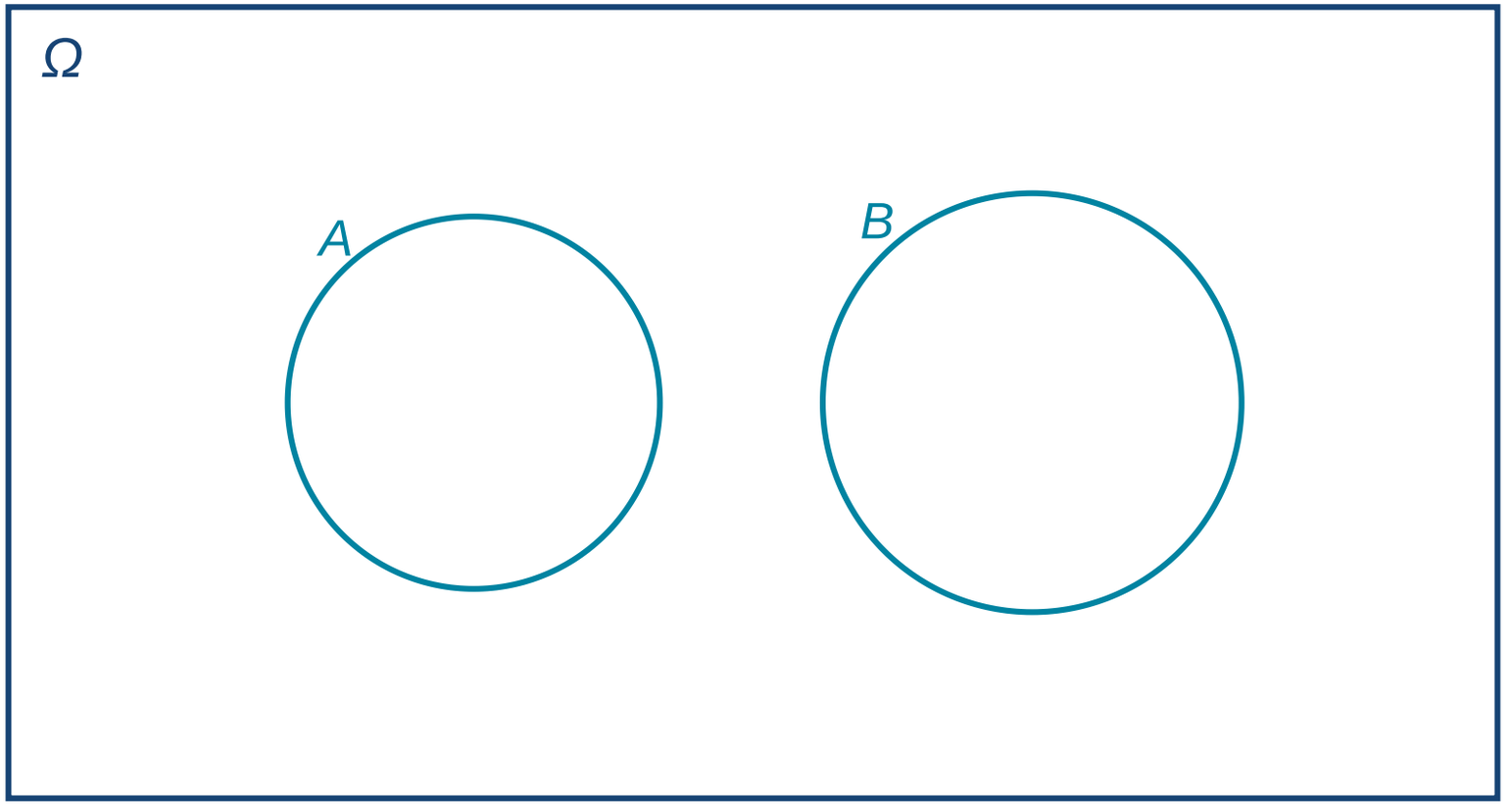
Speaker notes

Bridge from sets to pictures before any formula (P4). The payoff of the diagram: once probabilities are attached, probability laws become area arguments.

Important Event Types

Before each step: which region will light up?

$$A \cap B = \emptyset$$



A and B are disjoint

Speaker notes

Stepped visualization (P6, tier 2), one click per event type: the event A; the complement $\bar{A} = \Omega \setminus A$; the subset $B \subset A$; disjoint events $A \cap B = \emptyset$. Predict-then-check on every click — students name the region first, the highlight confirms. Disjointness is the one to slow down on: it is a statement about events, not about probabilities.

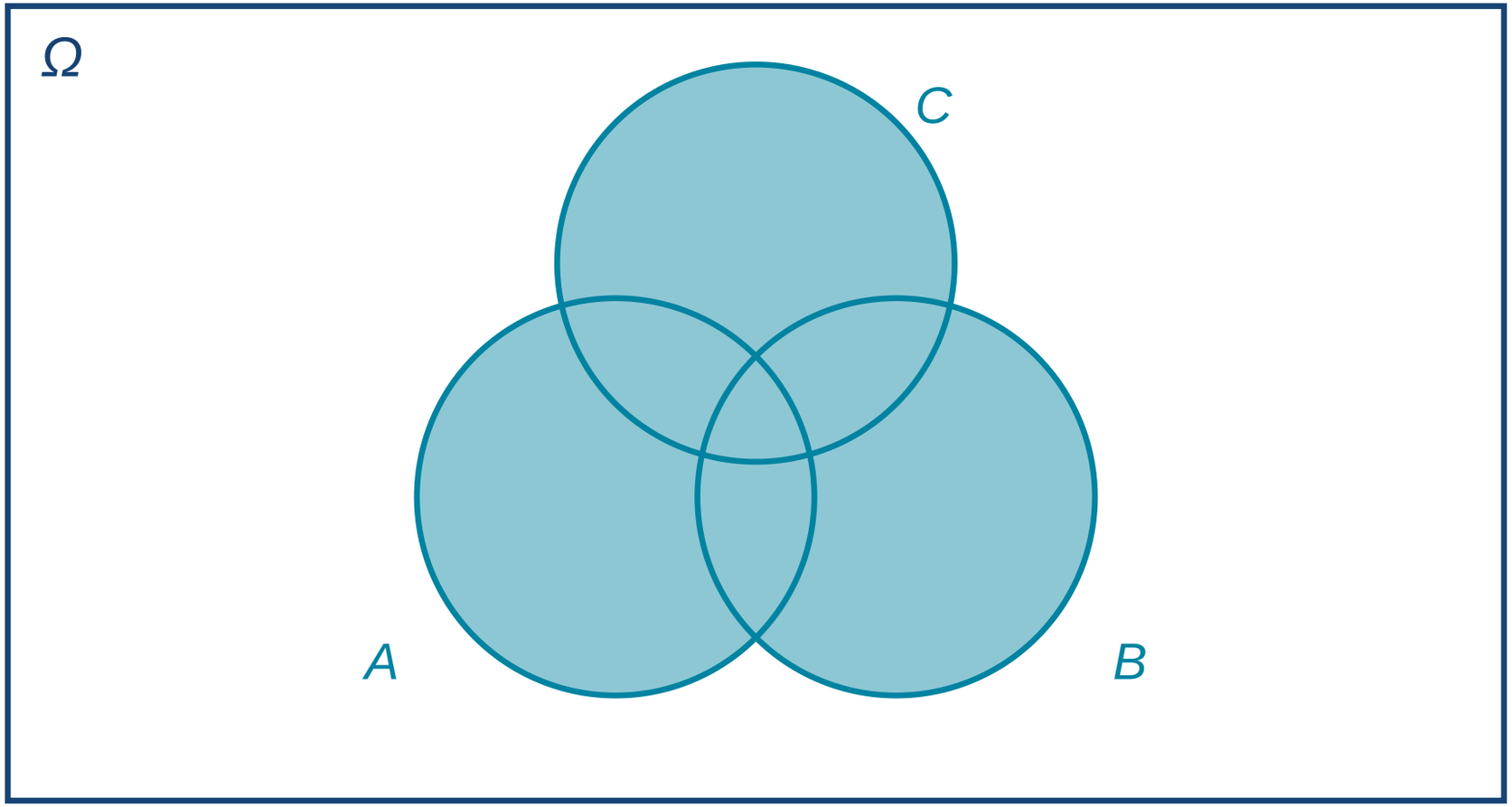
Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions

A ∪ B ∪ C



Union of three events

Speaker notes

Second stepped pass with three sets: A∪B, A∩B, A∩B∩C, A∪B∪C. Same predict-then-check rhythm.

Mini-Exercise

Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

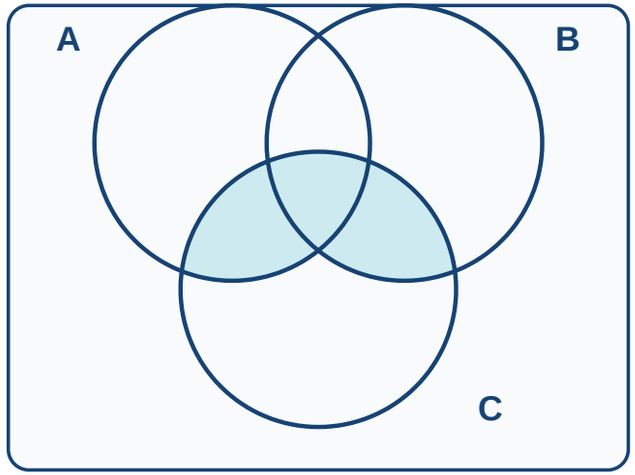
Speaker notes

Two minutes: students picture the regions in their head — which circle is the outer operation? — and commit to yes or no before the click. The figure is the answer; the algebra confirms it. If there is time, ask for a case where the two *are* equal (exactly when $A \cap B \subseteq C \subseteq A \cup B$).

Mini-Exercise

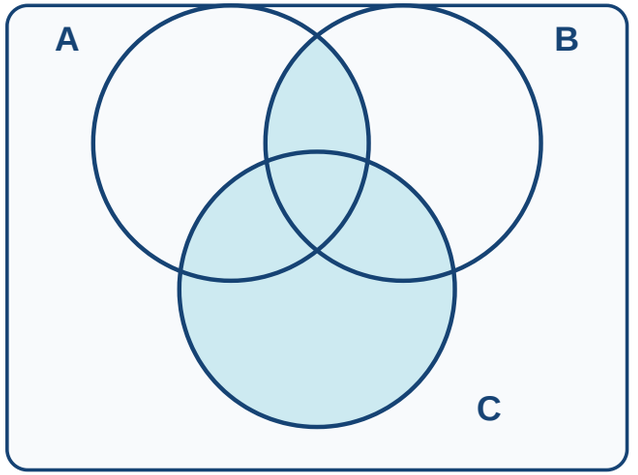
Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

(A ∪ B) ∩ C



inside C only

(A ∩ B) ∪ C



all of C, plus the lens A ∩ B

No. The first lies entirely inside C; the second is all of C plus the lens $A \cap B$. The distributive laws expand them into different expressions.

Takeaway

An event is a **subset** of the sample space, and combining statements about outcomes is set algebra: *or* is a union, *and* an intersection, *not* a complement. Choosing Ω decides which events exist at all.

Speaker notes

Module 1 closes here (course-local slide: the Venn unit ends on its exercise). The point to carry into Module 3: a probability will be a number attached to each of these subsets, so the algebra of events is what the rules of probability are written in.

Module 2

Counting

> ***Guiding Question:*** *How do we determine the size of a sample space we could never write down?*

Why Counting Comes First

If every outcome in a finite sample space is equally likely, then the probability of an event is a **ratio of counts**:

$$P(A) = \frac{|A|}{|\Omega|}$$

So before we can measure probability, we must be able to count $|A|$ and $|\Omega|$ — often without enumerating them.

Speaker notes

Motivate combinatorics as the arithmetic of sample spaces, not as a separate topic. Flag the assumption doing the work here — "equally likely" — which the axiomatic definition of probability replaces with an axiom system that does not need it.

Basic Principle of Counting

Generalised basic principle of counting. If an experiment consists of r stages, where stage i has n_i possible outcomes regardless of the earlier stages, then the whole experiment has

$$n_1 \cdot n_2 \cdots n_r$$

possible outcomes.

Example. How many 7-place licence plates, if the first 3 places are letters (26) and the last 4 are digits (10)?

$$26^3 \cdot 10^4 = 175{,}760{,}000$$

Speaker notes

The phrase "regardless of the earlier stages" is the hidden assumption — it fails as soon as choices interact (no repeated letters, say), which is exactly what the four cases below systematise.

Three Questions Decide the Formula

Order?

Does the sequence matter?
matters → **permutation** (rankings)
does not → **combination** (committees)

Repetition?

May an element be picked twice?
without — teams of people
with — letters in a word

How many?

Select k of n elements, or arrange all n?

Speaker notes

Give students a problem and have them answer these three questions *before* reaching for a formula. Nearly every counting error is a misread of one of these three, not an algebra slip.

The Four Basic Cases

Ordered, without repetition

k-permutation of n:

$$\frac{n!}{(n-k)!}$$

From $\{a, b\}$, $k = 2$: ab, ba

Ordered, with repetition

$$n^k$$

From $\{a, b\}$, $k = 2$: aa, ab, ba, bb

Unordered, without repetition

k-combination of n:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

From $\{a, b\}$, $k = 2$: $\{a, b\}$

Unordered, with repetition

$$\binom{n+k-1}{k}$$

From $\{a, b\}$, $k = 2$: $\{a, a\}, \{a, b\}, \{b, b\}$

Speaker notes

2x2 grid — the whole unit in one slide. Keep it visible while working examples; students should learn to locate a problem in this square rather than recall four formulas in isolation.

Permutations

Definition (k-permutation). An ordered selection of k elements from a set of n distinct elements without repetition. Counting position by position: n choices, then $n-1$, …, down to $n-k+1$, giving $\frac{n!}{(n-k)!}$.

Example. A class of 6 men and 4 women is ranked after an exam, no ties. How many possible rankings? $10! = 3{,}628{,}800$.

Speaker notes

The stage-by-stage argument *is* the basic counting principle applied with a shrinking pool — worth making explicit so the formula is derived, not given.

Combinations

Each k-combination corresponds to $k!$ different k-permutations — the same k elements in every possible order. Therefore

$$\binom{n}{k} = \frac{\text{number of } k\text{-permutations}}{k!} = \frac{n!}{k!(n-k)!}$$

Example. A committee of 3 from 20 people: $\binom{20}{3} = 1140$.

Dividing by $k!$ is the standard move for "forgetting" an ordering we counted.
Remember it as *over-count, then correct*.

Speaker notes

The over-count-then-divide pattern recurs constantly. Ask why we divide by k! and not by something else — because every unordered selection was counted exactly k! times, uniformly.

Properties of the Binomial Coefficient

$$\binom{n}{k} = \binom{n}{n-k}$$

Choosing which k to include is the same as choosing which $n-k$ to leave out.

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Split by a fixed element i: groups **containing** i, plus groups **not containing** i.

Binomial theorem.

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Speaker notes

Both identities are proved by counting the same set two ways — the standard combinatorial proof technique. The binomial theorem's proof is the same idea: expanding the product gives 2^n terms, and x^k y^(n-k) appears exactly as often as there are ways to pick which k factors contribute x.

Application: How Many Subsets?

A set with n elements has

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

subsets — sum over all subset sizes.

> **Alternative intuition.** Each element is either in the subset or not: a binary choice, n times, so $2 \cdot 2 \cdots 2 = 2^n$.

This is the size of the event space $\mathcal{P}(\Omega)$ from the sample-space unit.

Speaker notes

Close the loop with the sample-space unit: the 2^n we asserted there is now derived twice. Ask which of the two arguments they would trust more under pressure — the binary-choice one generalises better.

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Speaker notes

One point: the same selection, counted with and without order, differs by the factor $k!$ — over-count, then correct. Ask which cell of the four-case square each answer sits in.

Mini-Exercise

From 5 people, how many ways are there to pick a **chair and a deputy**? And how many ways to pick a **two-person committee**?

Chair and deputy: order matters, $5 \cdot 4 = 20$. Committee: each pair was counted twice (as ab and ba), so $20/2! = 10 = \binom{5}{2}$.

Takeaway

> Counting reduces to three questions — **order? repetition? how many?** — and one habit: when an ordering was counted that should not have been, divide it out.

Module 3

What Is a Probability?

> ***Guiding Question:*** *Every attempt to define probability turns out to be circular. What do we do instead?*

An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of A is how likely the stone lands inside A, assuming

1. we hit the rectangle Ω for certain, and
2. every part of Ω is equally likely to be hit.

Probability is then a **measure of the state** $A \subseteq \Omega$ of a random experiment — literally, how much of the sample space the event occupies.

Both assumptions are doing real work. Assumption 2 is the one we will have to give up.

Speaker notes

Intuition first (P4), and it connects directly to the Venn unit: events were regions, now regions have size. Ask what happens to the picture if some parts of Ω are "heavier" than others — that is precisely what the axioms will allow.

Two Ways to Read a Probability

Frequentist

The probability of an elementary event a is the fraction of repetitions of the experiment that yield a.

Requires the experiment to be repeatable.

Bayesian

The probability of a is the degree of certainty that a will happen — even without conducting the experiment.

Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

Speaker notes

Interpretation over recitation (P2): ask which reading makes sense for "the probability that this patient has the disease" (no repetition available) versus "the probability that this machine part fails within a year". The distinction returns whenever we call something a prior.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment**; its probabilities need not be estimated, they are implied by the setup.

Speaker notes

This is the definition that counting serves. Emphasise that it is only usable under the Laplace assumption; when that fails, probabilities must come from data.

Why That Definition Is Not Enough

> **The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

Speaker notes

This slide is the pivot of the module and answers the guiding question. Let the tension sit for a moment before the axioms arrive: students should feel why an axiomatic move is the honest way out, not a formal ritual.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \to \mathbb{R}$ is a **probability measure** if:

Axiom I

$$P(A) \geq 0$$

non-negative

Axiom II

$$P(\Omega) = 1$$

normalised

Axiom III

$$A \cap B = \emptyset \Rightarrow$$
$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

Beyond finite sample spaces. When Ω is infinite, the event space is not the full power set but a **σ-algebra** of measurable sets, and Axiom III is required for **countably many** disjoint events. Everything we compute in this course is unaffected; the construction belongs to measure theory.

Speaker notes

Math first (P5). Ask: which axiom does the area picture make obvious? (All three — areas are non-negative, the whole rectangle is everything, disjoint regions have additive areas.) The axioms are the area intuition, minus the requirement that the areas be equal.

Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

Speaker notes

Whenever a later theorem opens with "let (Ω, P(Ω), P) be a probability space", students should hear "we have agreed on outcomes, questions, and a measure".

What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

(Axioms II, III)

$$P(\emptyset) = 0$$

(from the above with $A = \Omega$)

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity *(Axioms I, III)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion *(Axiom III)*

For mutually exclusive $A_1, \ldots, A_p$: $P(A_1 \cup \ldots \cup A_p) = P(A_1) + \ldots + P(A_p)$.

Speaker notes

2x2 grid of consequences. Have students justify inclusion–exclusion on the Venn picture — the overlap was counted twice. Note that "A entails B implies A is no more likely than B" is monotonicity: the logic of events transfers to numbers.

Takeaway

> Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Speaker notes

Stabilise the epistemic move: an axiomatic definition says what the object must do, leaving where the numbers come from to the modeller. That freedom is what lets the same theory carry frequentist and Bayesian readings.

Laplace Probabilities in Practice

When all outcomes in Ω are equally likely, computing $P(A)$ means **counting twice**: the favourable cases, and the possible ones.

$$P(A) = \frac{|A|}{|\Omega|}$$

The whole difficulty is choosing a sample space in which "equally likely" is actually true.

Speaker notes

The formula is trivial; the modelling is not. Every error in the examples below comes from an Ω where outcomes are not equiprobable.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

Why ordered pairs? Because $(1, 6)$ and $(6, 1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Speaker notes

The classic trap. If a student proposes the 21 unordered pairs, ask them to bet on "double six" versus "one and six" — the intuition corrects itself fast.

Example: Sum of Two Dice

What is the probability that two dice sum to 7?

- $|\Omega| = 6 \cdot 6 = 36$ ordered pairs
- $E = \{(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)\}$, so $|E| = 6$

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Why ordered pairs? Because $(1,6)$ and $(6,1)$ are two distinct equally likely outcomes — the unordered sample space is **not** equiprobable.

Example: Ordered or Unordered — Both Work

An urn holds **6 white** and **5 black** balls. Three are drawn. **What is the probability of exactly 1 white and 2 black?**

Variant 1 — order matters

$|\Omega| = 11 \cdot 10 \cdot 9 = 990$

Favourable: white-black-black, black-white-black, black-black-white $= 6 \cdot 5 \cdot 4 + 5 \cdot 6 \cdot 4 + 5 \cdot 4 \cdot 6 = 360$

$$\frac{360}{990} = \frac{4}{11}$$

Variant 2 — order ignored

$|\Omega| = \binom{11}{3} = 165$

Favourable: $\underbrace{\binom{6}{1}}_{\text{the white}} \cdot \underbrace{\binom{5}{2}}_{\text{the blacks}} = 6 \cdot 10 = 60$

$$\frac{60}{165} = \frac{4}{11}$$

Speaker notes

Same answer, two sample spaces — the point of the slide. Both are legitimate *because* both are equiprobable: ordered draws are equally likely, and so are unordered groups. Mixing the two (favourable counted unordered, total counted ordered) is the error to warn against.

The General Rule

> When selecting k items from n, you may count **with** or **without** order — as long as you are consistent, and as long as the chosen sample space really is equiprobable.

Pick whichever counting is easier for the event at hand.

Speaker notes

Connects straight back to the four cases in the counting unit: the choice of case is a modelling choice about Ω, not a property of the problem.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Speaker notes

One point: either counting works if it is used consistently. The common error to listen for is 4 favourable unordered pairs over 12 ordered draws (1/3). The earlier ten-couples problem ("five chosen from 10 couples, none a couple", $\binom{10}{5}2^5/\binom{20}{5} \approx 0.52$) belongs on the exercise sheet.

Mini-Exercise

The urn again, smaller: **2 white and 2 black** balls, two drawn. What is the probability of **one of each**? Count without order, then check with order.

Unordered: $\binom{4}{2} = 6$ pairs, $2 \cdot 2 = 4$ mixed, so $4/6 = 2/3$. Ordered: $4 \cdot 3 = 12$ draws, $2 \cdot 2 + 2 \cdot 2 = 8$ mixed, so $8/12 = 2/3$ — the same, because both sample spaces are equiprobable.

Takeaway

Laplace's formula — favourable over possible cases — is **one** probability measure, valid only in a sample space whose outcomes are equally likely. The formula is trivial; choosing that sample space is the work.

Speaker notes

Module 3 closes here (course-local slide: the examples unit ends on its exercise). Connect back to the axioms: Laplace satisfies them, which is why it is a probability — not the other way round.

Module 4

Conditioning, Independence, Bayes

> ***Guiding Question:*** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what is the correct bookkeeping?*

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

Speaker notes

Intuition first (P4): conditioning is not a new kind of probability, it is the old probability computed in a **smaller sample space**. Let students do the count before the definition appears.

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll: $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3,1),\ldots,(3,6)\}$ remains
- Outcomes like $(2,4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3,5)\}$

$$P = \frac{1}{6}$$

Definition

Definition (Conditional probability).

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{if } P(B) > 0$$

$P(A \mid B)$ is the probability of A **under the condition** B.

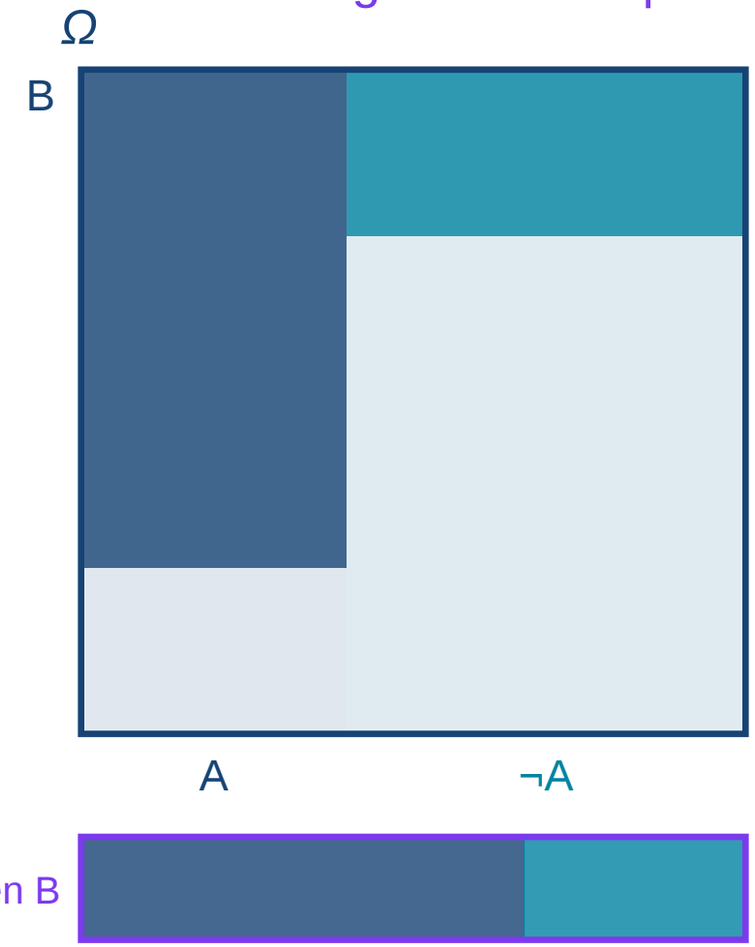
Back to the stone: *given that the stone landed in area B, how likely is it that it also landed in A?* We renormalise by the area of B, because B is now the whole world.

Speaker notes

Math first (P5) after the dice intuition. The division by P(B) is exactly the renormalisation students performed implicitly when they used the six remaining outcomes.

Conditioning, Drawn

Conditioning on B = keep the B cells, stretch them back to width 1



P(A) = 0.40

P(B | A) = 0.75, P(B | ¬A) = 0.25

P(B) = 0.45

P(A ∩ B) = 0.30

P(A | B) = 0.67

Before the last click: what must the two shaded cells be divided by?

Speaker notes

Stepped visualization (P6): the A columns; B's share inside each column; keep only B; renormalise. The stretch factor *is* the 1/P(B) in the definition, so the picture performs the formula rather than illustrating it.

Then use the sliders. Two settings are worth doing live: drag P(B|A) and P(B|¬A) to the same value — the bar's split stops moving and P(A|B) = P(A), which is independence, met before it has a name. Then set P(B|¬A) near zero: B becomes nearly conclusive evidence for A. Ask what has to be true of the two conditionals for evidence to be worth anything.

Example: Two Coins

Two coins are flipped. The event of interest is $B = \{(h, h)\}$.

(a) "The first coin is heads"

$F = \{(h, h), (h, t)\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$A = \{(h, h), (h, t), (t, h)\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event B, two different pieces of information, two different answers. **Why is (b) smaller?**

Speaker notes

Force interpretation (P2). The information in (a) is stronger — it names *which* coin — so it eliminates more of the sample space. This is the classic "boy or girl" puzzle in disguise; expect disagreement, which is the point.

Total Probability

Theorem (Total probability). Let $B_1, \ldots, B_k$ be mutually exclusive with $\Omega = B_1 \cup \ldots \cup B_k$ and $P(B_i) > 0$. Then for any event A:

$$P(A) = \sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)$$

Proof. Decompose A along the partition and use additivity:

$$P(A) = P((B_1 \cup \ldots \cup B_k) \cap A) = \sum_{i=1}^{k} P(B_i \cap A) = \sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)$$

Speaker notes

Two axioms do all the work: completeness of the partition, then exclusiveness for additivity. This theorem is the denominator of Bayes' rule — say so now, so that Bayes looks assembled rather than conjured.

Rules That Follow

Product rule

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$
$$= P(A) \cdot P(B \mid A)$$

Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B, the map $A \mapsto P(A \mid B)$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B.

Speaker notes

2x2 grid. The top-right box is already Bayes' theorem — flag it, then move on; the Bayes unit gives it its name and its interpretation. The bottom two boxes are the most common exam mistake: you may complement the *event*, not the *condition*.

Chain Rule

$$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \ldots \cap E_{n-1})$$

By successive application of the product rule.

> Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Speaker notes

Do not belabour the proof; the payoff is the remark. The chain rule plus conditional independence is the entire idea behind belief networks.

Takeaway

> Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(A \mid B) = P(A \mid \overline{B})$$

Independence means: **learning B changes nothing about A.**

Speaker notes

The conditional form is the meaningful one; the product form is the one that is easy to check and stays defined when P(B) = 0. Emphasise the reading: knowing B tells you nothing new.

The Classic Confusion

> **Statistical independence does not mean the events are disjoint.**

Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

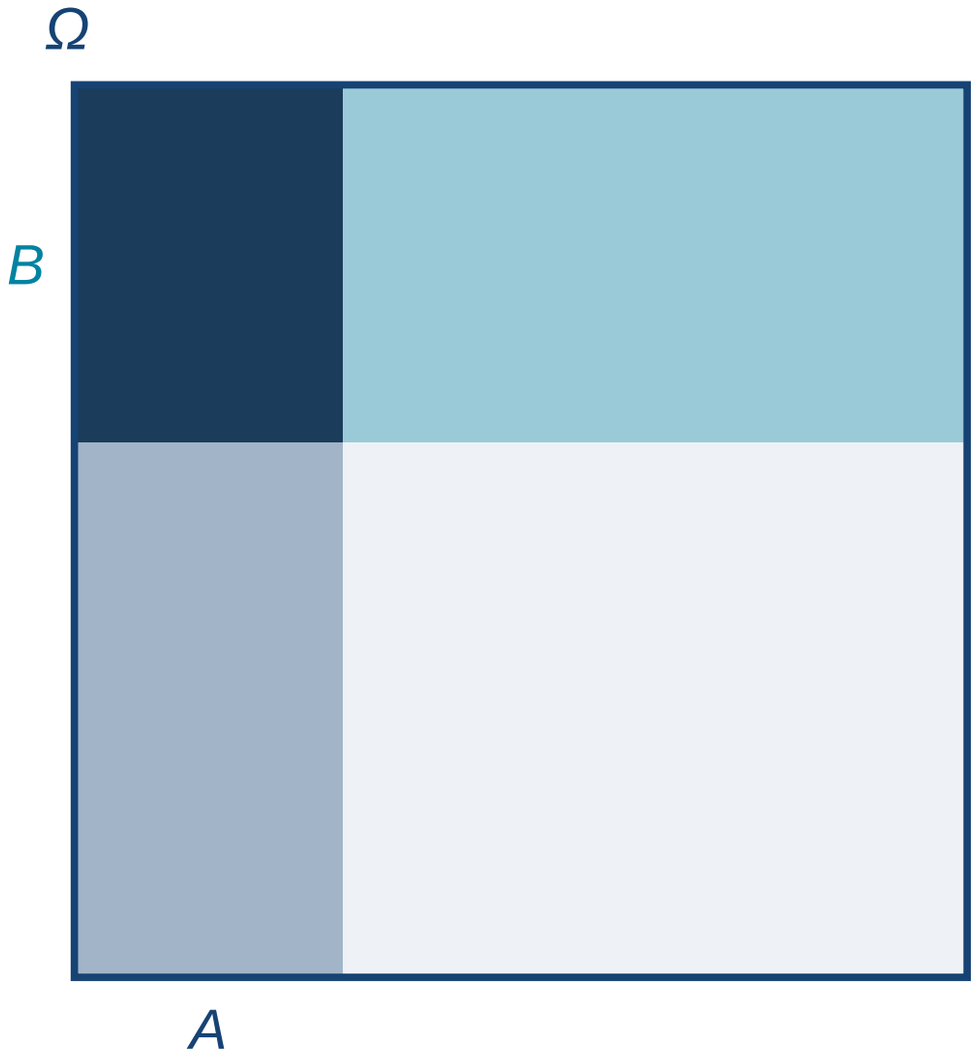
$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

Speaker notes

Ask students to state, in one sentence, why disjointness is the opposite of independence. The Venn picture misleads here — non-overlapping circles look "unrelated", which is exactly wrong.

Independence Is About Areas, Not Overlap

$0.30 \cdot 0.40 = 0.12 = P(A \cap B) \rightarrow$ independent



P(A) = 0.30

P(B) = 0.40

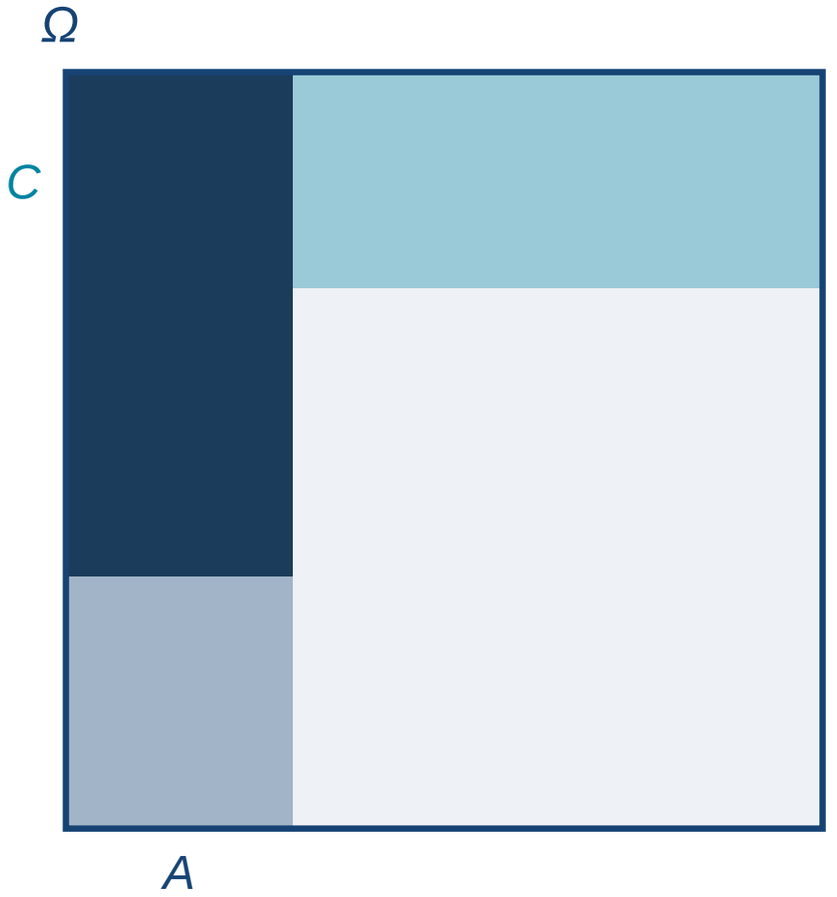
P(A ∩ B) = 0.12

Speaker notes

Stepped viz (P6): band A, band B, their intersection, then the test. $0.3 \cdot 0.4 = 0.12$ — independent. Ask before the last click whether they expect equality. The lesson: independence is a numerical coincidence of areas, not a visual property of the drawing.

The Same Picture, Dependent

$0.30 \cdot 0.40 = 0.12 \neq 0.20 = P(A \cap C)$ → dependent



P(A) = 0.30

P(C) = 0.40

P(A ∩ C) = 0.20

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

Speaker notes

Same marginals as the previous slide, different joint. This is the payoff of the pair: the marginals never tell you whether events are independent. Later, that is exactly why joint distributions are needed.

More Than Two Events

Events $A_1, \ldots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \ldots, i_l$:

$$P(A_{i_1} \cap \ldots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Speaker notes

Give the standard counterexample if time allows (two fair coins and the event "the results agree" are pairwise independent but not mutually independent). The distinction matters for naive Bayes, which assumes far more than pairwise.

Takeaway

> Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Conditional Independence

Definition. Events A and B are **conditionally independent given** C (written $A \perp\!\!\!\perp B \mid C$) iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

equivalently

$$P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B carries no further information about A.**

Speaker notes

Math first (P5), but immediately give the reading. The second form is the one used in practice: it says B becomes redundant in the presence of C.

Dependent, but Independent Given C

Let $P(A) = \frac{8}{16}$, $P(B) = \frac{3}{16}$, $P(C) = \frac{4}{16}$ on a 16-cell sample space.

Unconditionally dependent

$$P(A \cap B) = \frac{1}{16}$$

$$P(A) \cdot P(B) = \frac{3}{32}$$

$\frac{1}{16} \neq \frac{3}{32}$, so $A \not\perp\!\!\!\perp B$

Conditionally independent given C

$$P(A \cap B \mid C) = \frac{1}{4}$$

$$P(A \mid C) \cdot P(B \mid C) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$$

so $A \perp\!\!\!\perp B \mid C$

Also directly: $P(A \mid B \cap C) = \frac{1}{2} = P(A \mid C)$.

Speaker notes

The whole idea in one comparison. Conditioning on C *created* independence that did not exist before. Ask for a real example: ice-cream sales and drownings are dependent, but conditionally independent given the temperature.

Why This Matters

Common cause
A common cause C can make two effects look related; given C, they are independent.

Common effect
Two independent causes become dependent once their common effect is observed — *explaining away*.

Model size
Conditional independence lets a joint distribution factorise into small pieces — naive Bayes and graphical models are built on it.

Conditional independence neither implies nor is implied by ordinary independence. Both directions can fail.

Speaker notes

The first two cards are the two directions of the takeaway: conditioning on a common cause *removes* a dependence (ice cream and drownings, given the temperature); conditioning on a common effect *creates* one (a wet lawn: rain and the sprinkler, once you know the grass is wet, compete to explain it). Name the models that rest on the definition without developing them. The warning box is worth a moment — students routinely assume one form gives the other.

Takeaway

> Conditioning can **create** independence as easily as it destroys it. "Given what I already know, does this extra fact still tell me anything?" is the question that makes large probabilistic models tractable.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Speaker notes

Intuition-first entry (P4). Collect guesses by show of hands *before* clicking: most say 99%. The surprise is the motivation for the whole module. Do not explain yet — the next slide lets them count it themselves.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

> Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

P(sick | positive) = 1/11 ≈ 9%



Speaker notes

Stepped visualization (P6 tier 2), one click per step: the population; the one sick person; the ~10 false alarms among the 999 healthy; restricting to the 11 positives. Ask before the last click: "of these 11 flagged people, how many are sick?" Everything here is counting and restriction of the sample space — no new machinery. This *is* conditioning, drawn.

What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B; the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Speaker notes

Name the confusion explicitly — swapping the two conditionals is the single most common probabilistic error, in exams and in the newspapers. Bayes' theorem is the machine that converts one into the other.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

Speaker notes

Math-first-then-meaning (P5), legitimate here because the intuition already exists. Stepped equation walk-through: each click highlights one part and its brace — posterior, likelihood, prior, evidence — while the rest dims. Map the disease test onto the symbols as you go: A = sick, B = positive test. P(A) = 1/1000 is the term everyone forgets. Reference: (Sheldon & others 2002).

Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Speaker notes

One line: expand, multiply by P(A)/P(A), recognise the likelihood. Worth asking which step needed P(B) > 0 and which needed P(A) > 0.

Bayes' Theorem — Multiple Hypotheses

Theorem (Bayes, multiple events). Let $A_i \in \mathcal{P}(\Omega)$ be mutually exclusive with $\Omega = A_1 \cup \ldots \cup A_k$ and $P(A_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(A_i \mid B) = \frac{P(B \mid A_i) \cdot P(A_i)}{P(B)} = \frac{P(B \mid A_i) \cdot P(A_i)}{\sum_{j=1}^{k} P(B \mid A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

Speaker notes

The hypotheses partition Ω — "sick vs. healthy" was the k = 2 case. This is the form that reappears as a classifier (naive Bayes) and in belief networks: score every hypothesis by likelihood × prior, then normalise.

Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j) \cdot P(A_j)$$

Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate $P(\text{sick} \mid \text{positive})$ before computing it.

Mini-Exercise — Solution

Speaker notes

Solution fragmented per the animation policy. The punchline is the comparison with 9%: identical likelihoods, different prior, completely different clinical meaning.

Mini-Exercise — Solution

Out of 1,000: 50 sick → 49.5 true positives; 950 healthy → 9.5 false alarms.

$$P(\text{sick} \mid \text{positive}) = \frac{49.5}{49.5 + 9.5} = \frac{49.5}{59} \approx 84\%$$

The test did not change. The prior did. This is why screening decisions depend on who is screened.

Takeaway

> Bayes' theorem is the arithmetic of changing your mind: **posterior ∝ likelihood × prior**. Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Speaker notes

Stabilise both halves: the base-rate trap (9%) and the prior's power (84%). Both return in Bayesian decision making and graphical models.

Module 5

Random Variables and Distributions

> ***Guiding Question:*** *We rarely care which outcome occurred, only what it was worth. How do we move from outcomes to quantities?*

Speaker notes

Session boundary: the second session starts here. Recall the three-card flow from the opening (assume → observe → check): Modules 1–4 built the model of chance for events; Modules 5–7 turn outcomes into quantities, describe a sample and check a difference.

Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins: $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$, eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads** Y:

$P(Y = 0) = \frac{1}{8}$	$P(Y = 1) = \frac{3}{8}$	$P(Y = 2) = \frac{3}{8}$	$P(Y = 3) = \frac{1}{8}$
(T, T, T)	$(H, T, T), (T, H, T), (T, T, H)$	$(H, H, T), (T, H, H), (H, T, H)$	(H, H, H)

Y compresses eight outcomes into four numbers — and the probabilities come along automatically.

Speaker notes

Intuition first (P4). Ask what was lost in the compression (which coin showed heads) and what was gained (a quantity we can average, compare, plot). That trade is the entire point of a random variable.

Definition

Definition (Random variable). A random variable X maps outcomes to a measurable range E: $X : \Omega \to E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

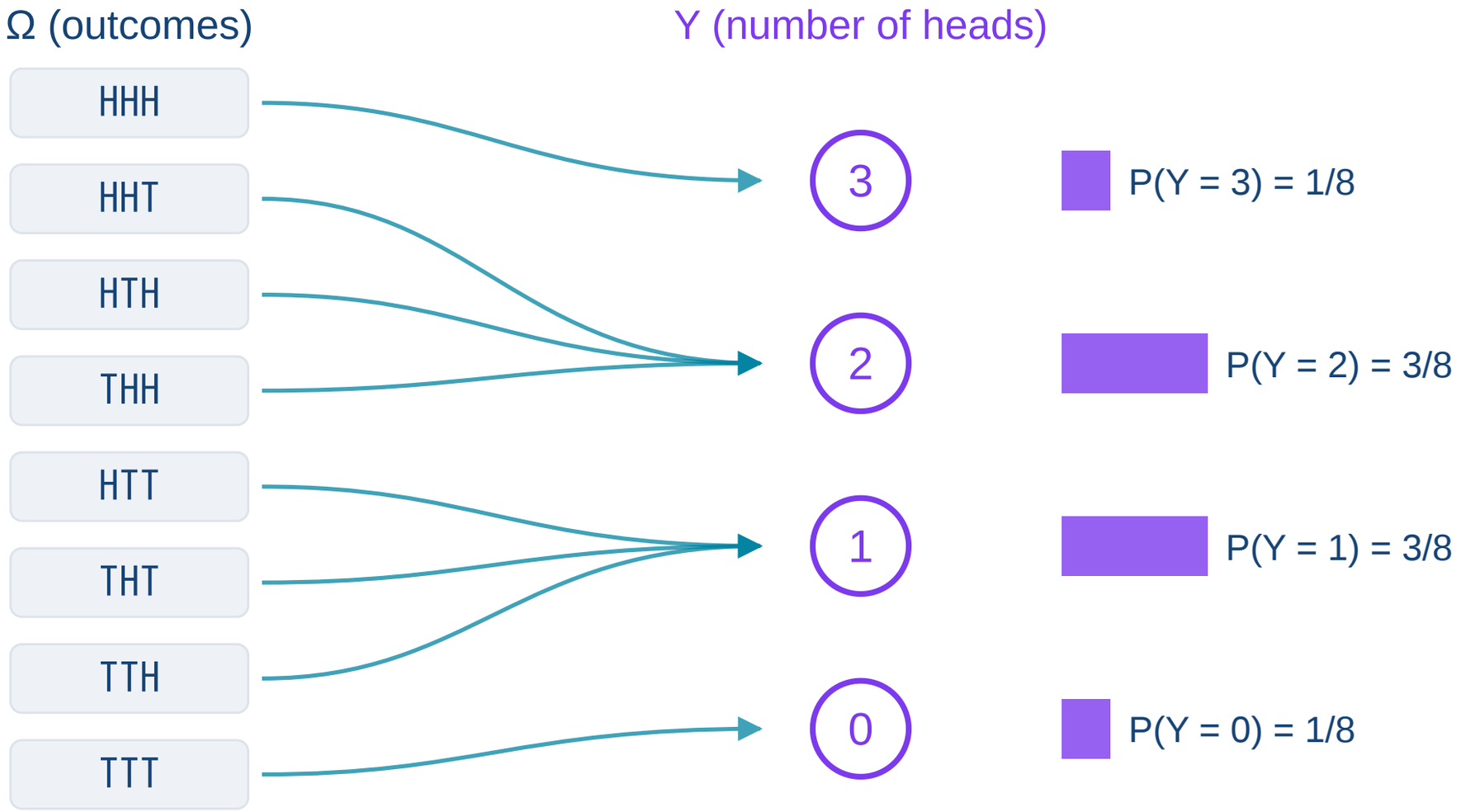
A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω, not in X.

Speaker notes

The naming trap is worth one minute; it removes a persistent confusion. The formula says: to get the probability of a value, collect all outcomes that map to it and use the measure we already have.

The Function, Drawn

Three arrows arrive at 1 and at 2 — which is the whole shape



Speaker notes

Stepped visualization (P6), three coins: the eight outcomes; the arrows, one per outcome; the four values with their counts; the resulting mass function. Two things to say out loud while stepping. First, *every* outcome has exactly one arrow leaving it — that is what makes Y a function. Second, three arrows arrive at 1 and three at 2, and that is the entire reason those values are more probable. The probability was never invented; it was inherited.

Examples

Two coins

$\Omega = \{HH, HT, TH, TT\}$

X = number of heads

People

$\Omega = \{\text{person}_1, \text{person}_2, \ldots\}$

X = height of the person

Two dice

$\Omega = \{\langle n_1, n_2 \rangle\}$

$X(\omega) = n_1 + n_2$

Speaker notes

The middle example is the one that connects to data science: a dataset column *is* a random variable — a function from records to values. Say it explicitly; it makes the next chapters feel continuous with this one.

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — with a mass function or with a density.

Speaker notes

Keep this light: the deep measure-theoretic reasons are out of scope. What students need is the practical consequence — sums for discrete, integrals for continuous.

A Random Variable With Infinitely Many Values

Flip a biased coin ($P(H) = p$) until the first head. Let X be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1-p)p, \quad P(X = 3) = (1-p)^2 p$$

$$P(X = n) = (1-p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

Speaker notes

This is the geometric distribution, met before it is named — fine, because it is constructed from the chain rule the students already have. Ask them to verify the sum is 1 (geometric series) if the group is comfortable with it.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X, the **probability mass function** is

$$p(a) = P(X = a)$$

$p(a) > 0$ for at most countably many a

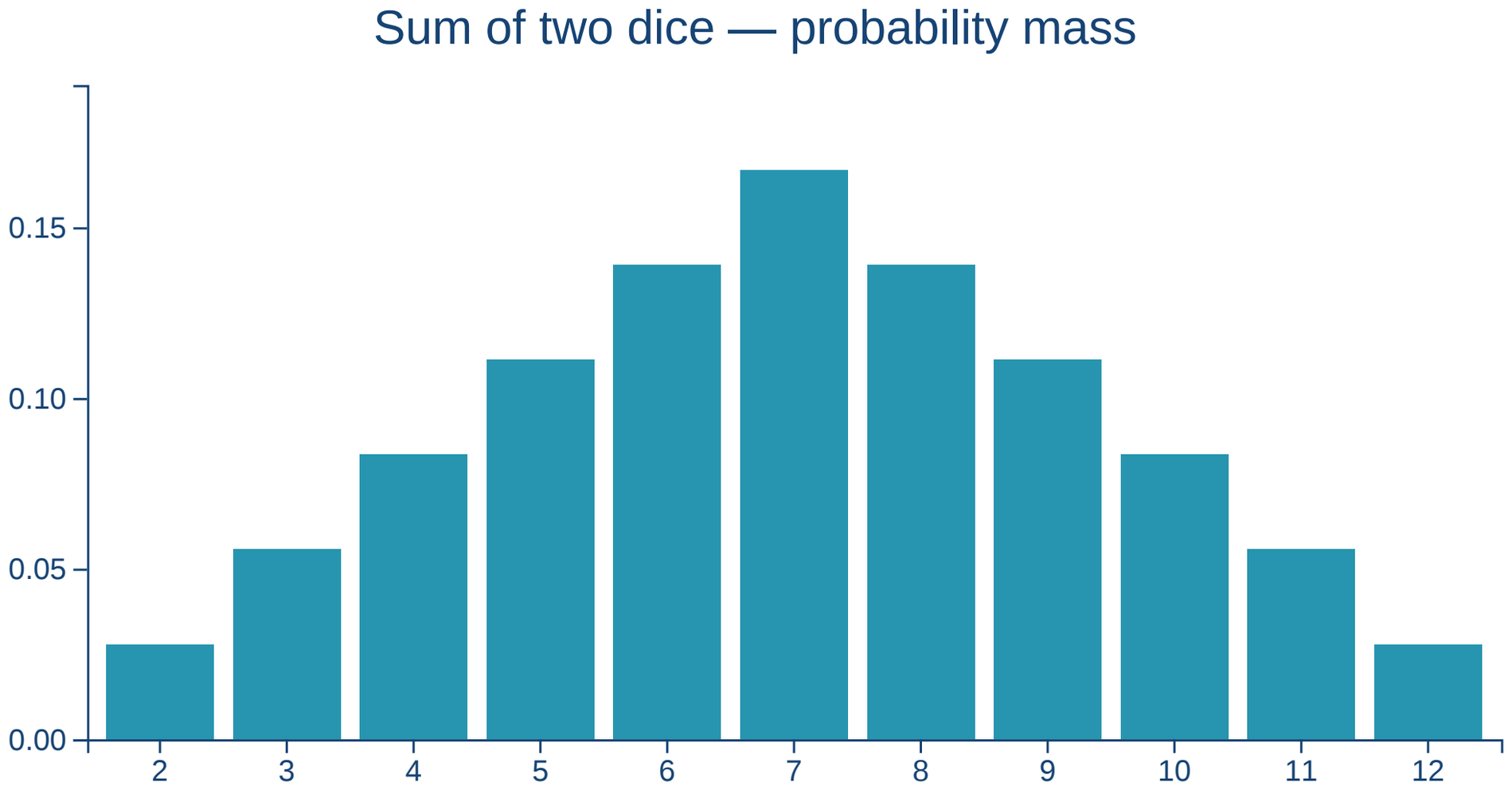
$\sum_i p(a_i) = 1$

p is itself a probability measure

Speaker notes

The PMF is just the probability measure, re-indexed by values instead of outcomes. Everything from the axioms still applies — which is why the three properties are consequences, not new assumptions.

Example: Sum of Two Dice



The triangular shape is counting, drawn: six of the 36 outcomes give 7, only one gives 2.

Speaker notes

Connect the picture to the Laplace assumption: 36 equally likely outcomes, counted by sum. Ask why the shape is symmetric and peaked — more ways to make a middling sum. This is the first hint of why sums of random quantities concentrate.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment: $\mu = E[X^1]$; the first *central* moment $E[X - E[X]]$ is always 0

For a real function g:

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants a, b:

$$E[aX + b] = aE[X] + b$$

Speaker notes

The linearity identity is the one used constantly later (in regression, in evaluation). Note that $E[g(X)] \neq g(E[X])$ in general — ask for a counterexample with $g(x) = x^2$.

Variance

Definition (Variance).

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$
- $\sigma = \sqrt{\mathrm{Var}(X)}$ is the **standard deviation**, in the same unit as X

Speaker notes

Two things to stabilise: the computational identity $E[X^2] - E[X]^2$, and why we report σ rather than the variance (units). The b disappearing under variance is worth a question: shifting data does not change its spread.

Takeaway

A discrete distribution is a PMF; expectation and variance compress it into **where it sits** and **how much it spreads**.

Bernoulli and Binomial

Bernoulli(p) — one trial

$X = 1$ success, $X = 0$ failure

$$E[X] = p, \quad \mathrm{Var}(X) = p(1-p)$$

Binomial $B(n, p)$ — n independent trials

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

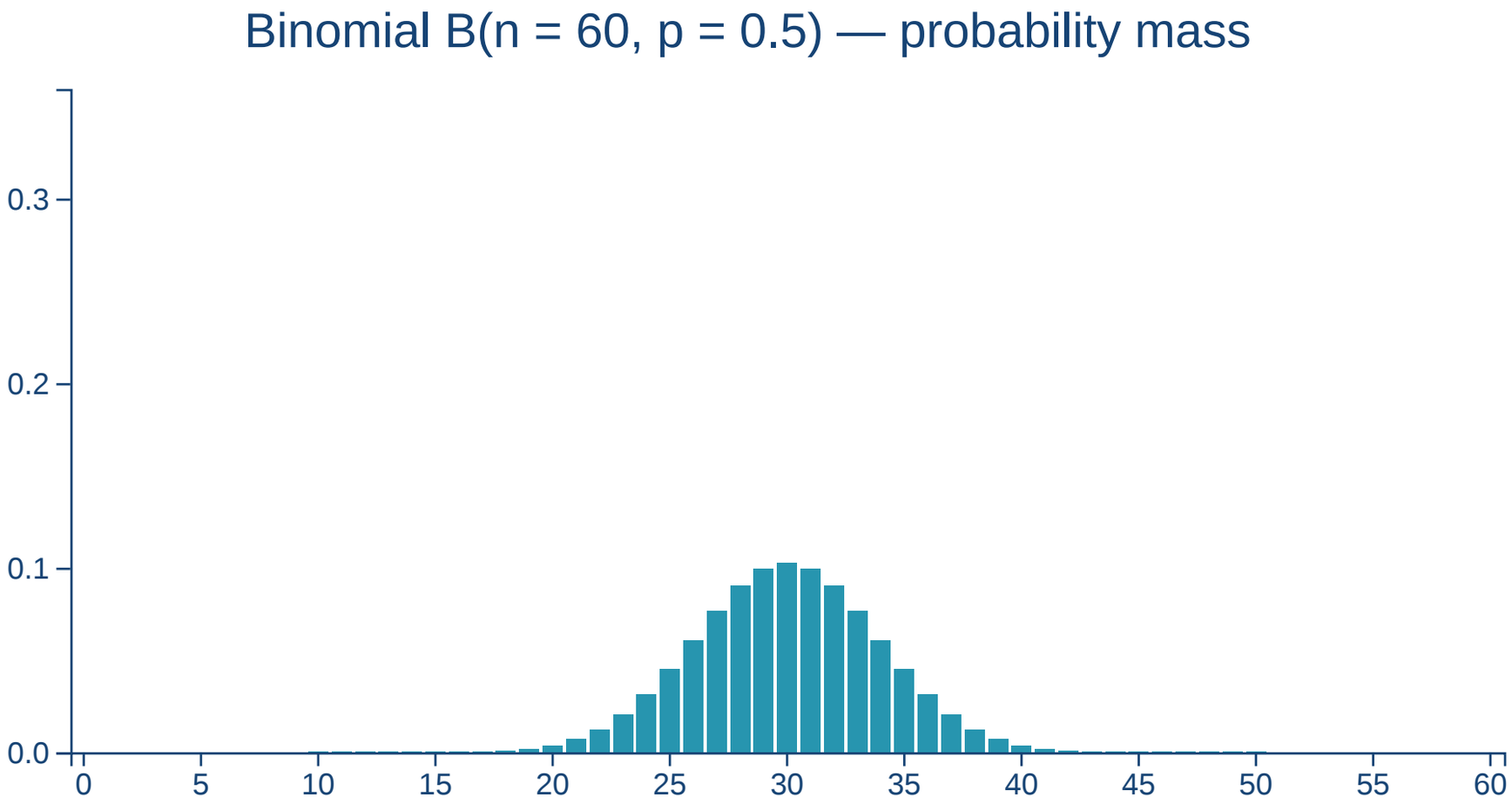
$$E[X] = np, \quad \mathrm{Var}(X) = np(1-p)$$

$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$ counts the sequences with k successes; the PMF multiplies it by the probability of one such sequence — combinatorics and independence in a single line.

Speaker notes

Point at where each ingredient comes from: the binomial coefficient counts the ways to place k successes among n trials, the product p^k(1−p)^(n−k) comes from independence. Nothing here is new machinery, which is exactly the message.

How the Shape Depends on the Parameters



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Speaker notes

Predict-then-check (P6). Two things happen at once: the spread in absolute terms grows (√n) while the shape becomes symmetric and bell-like. The second observation is the central limit theorem at work — a sum of many independent coin flips. Name it as an observation; the theorem itself is stated with the normal distribution.

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

Speaker notes

Two minutes: the expectation is head arithmetic, the second half only asks for a judgement. If someone wants the exact tail, it is $P(X \geq 5) \approx 0.078$ — a sum over six PMF values, not one. The modelling point matters more than the number: the binomial needs independent guesses with a constant success probability; ask when that would fail in a real exam.

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

$B(10, 0.25)$: on average $10 \cdot 0.25 = 2.5$ correct. Five or more is twice the expectation — unlikely (about 8%).

Takeaway

A yes/no attribute is a **Bernoulli** variable with one parameter p; counting the yeses in n independent trials gives a **binomial**. Both are counting in probabilistic clothing — and p is the number data will have to tell us.

From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

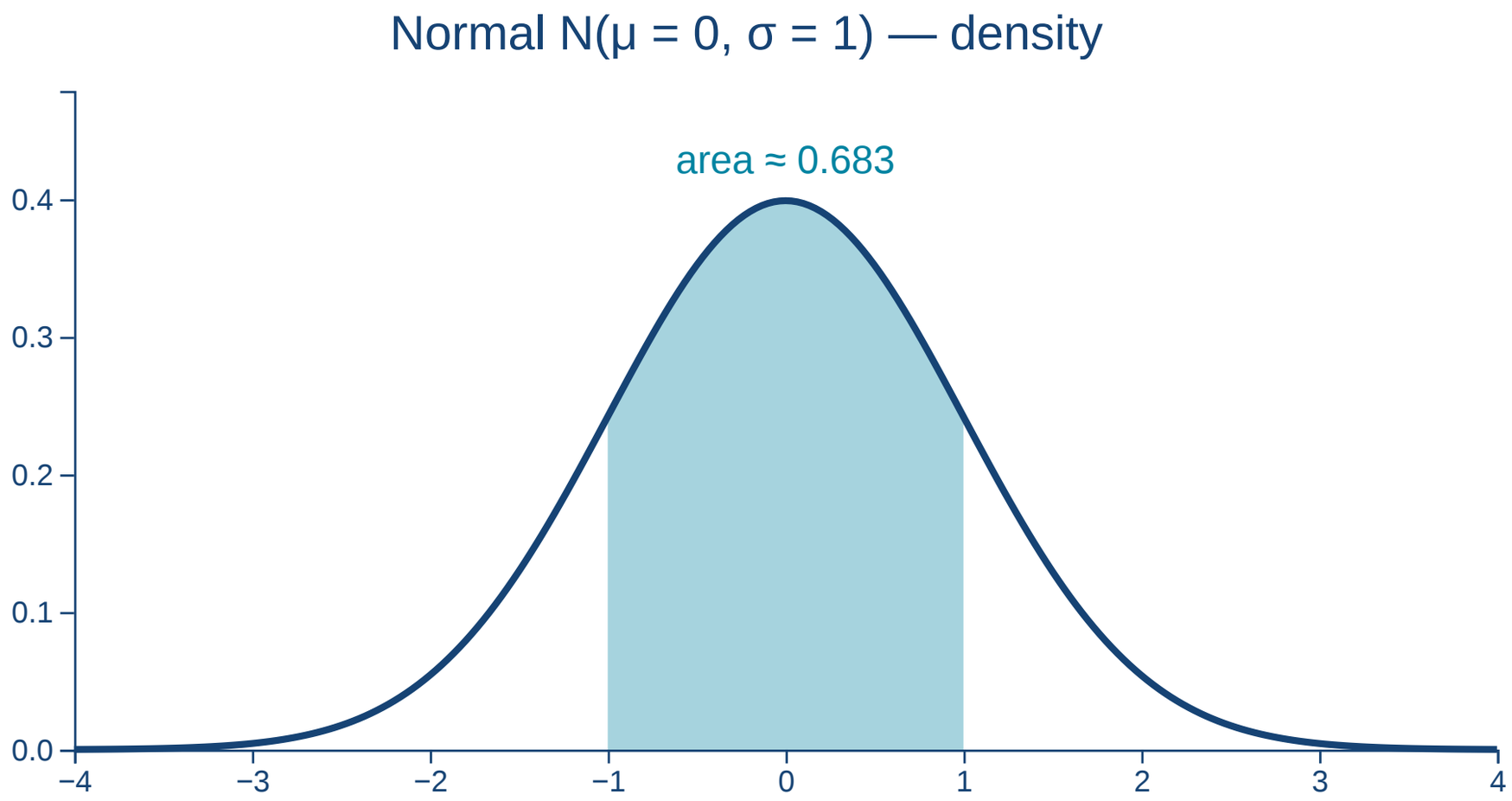
$$P(b \leq X \leq c) = \int_b^c f(a)da, \qquad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a.

Speaker notes

The density-is-not-probability point is the single biggest source of confusion in this unit. A useful question: what is the probability that a person is exactly 180.000... cm tall? Zero — probability lives on intervals now.

Probability Is Area Under the Curve



The shaded region is $P(-1 \leq X \leq 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Speaker notes

Deliberate callback (P7 spiral): area was the intuition for the axioms, and it still is. Ask what the total area must be, and why that forces the normalisation constant in the density formula.

Expectation and Variance — Sums Become Integrals

Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged: $E[aX + b] = aE[X] + b$, and $\mathrm{Var}(X) = E[(X - \mu)^2]$.

Speaker notes

Emphasise the continuity of concepts across the discrete/continuous divide: only the summation symbol changed. Students who see this stop treating the two cases as separate theories.

Two Basic Continuous Distributions

Uniform on $[a, b]$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

Every interval of equal length is equally likely — the Laplace assumption, continuous version.

Normal $N(\mu, \sigma^2)$

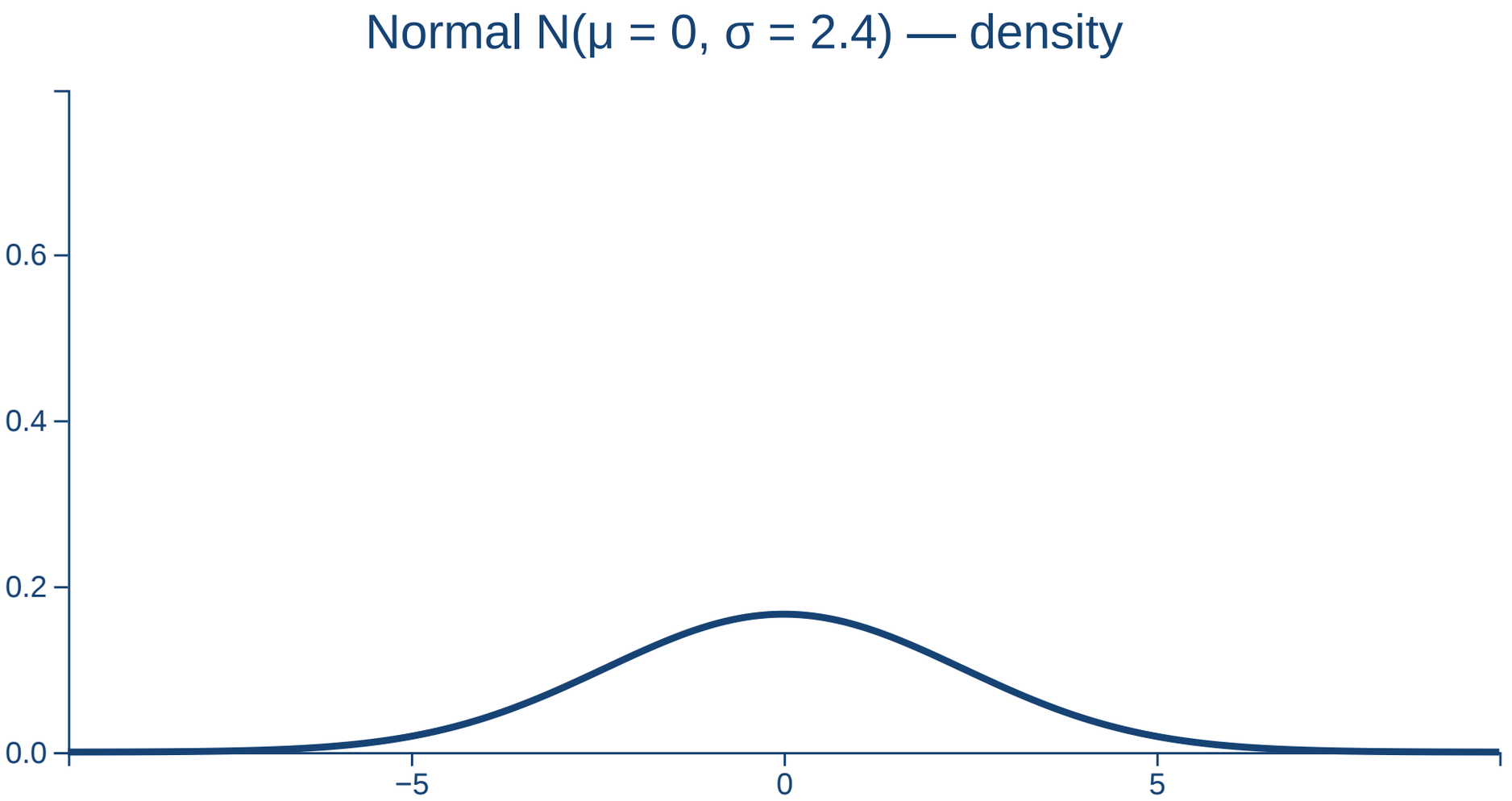
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

Speaker notes

The uniform is worth naming as the continuous Laplace assumption — it makes the earlier discussion feel like one idea rather than two.

What σ Does to the Shape



The peak falls as the curve widens — the area is always 1.

Speaker notes

Predict-then-check: ask what must happen to the height when the width grows. The constraint "total area = 1" is doing the work, and it is the same normalisation axiom that defined a probability measure.

Why the Normal Is Everywhere

> The **central limit theorem**: under mild conditions, the average of n independent observations is approximately normal for large n — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown — and why a sum of many independent coin flips, the binomial, looks bell-shaped.

Speaker notes

State the CLT, do not prove it. If the binomial sweep has already been shown, call back to it — students already saw it happen; if it comes later, it will confirm the theorem. This also sets up the sampling distribution of a mean, which every error bar computed from a sample rests on.

Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until n events have occurred
- **Beta** — values in $[0, 1]$, e.g. an unknown probability
- **Dirichlet** — distributions *over* probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

Speaker notes

The last two (Beta, Dirichlet) are named for recognition only — they return wherever a probability is itself unknown, as priors over probabilities. That is a legitimate forward *pointer*, not a forward dependency: nothing here uses them.

More Than One Random Variable

Univariate	Bivariate	Multivariate
one random variable	two random variables	three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Speaker notes

Make the dataset connection explicit: a table with columns age, income, and churn is a sample from a trivariate joint distribution. Everything in the rest of the course operates on joints.

Joint Mass and Density

Discrete — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

Continuous — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b)dadb$$

Notational convention from here on. We stop distinguishing discrete from continuous variables and simply write $p(a,b,c,\ldots)$ for the probability that $\{X_1, X_2, X_3, \ldots\}$ is in state $a, b, c, \ldots$

Speaker notes

Flag the convention as a deliberate simplification, and say why it is safe: the rules (marginalise, condition, factorise) are identical; only the mechanics of summing versus integrating differ.

Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \ldots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \ldots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

Speaker notes

Marginalisation is the total-probability theorem, applied to random variables — say so, since students met the theorem with conditional probability. That continuity is worth more than the formula.

Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability $\frac{1}{2}$.

Let B = number of boys, G = number of girls.

$p(B, G)$	$G = 0$	$G = 1$	$G = 2$	$G = 3$	$p(B)$
$B = 0$	.15	.10	.0875	.0375	.375
$B = 1$	.10	.175	.1125	0	.3875
$B = 2$	.0875	.1125	0	0	.2
$B = 3$	.0375	0	0	0	.0375
$p(G)$	.375	.3875	.2	.0375	1

The margins of the table *are* the marginals — which is where the name comes from.

Speaker notes

Have students read one cell aloud as a sentence ("11.25% of families have one boy and two girls"). Then ask whether B and G are independent — they are not, since the family size couples them. Good preparation for covariance.

Covariance

$$\operatorname{cov}(A, B) = E\big[(A - E[A])(B - E[B])\big]$$

$\operatorname{cov} > 0$
A large when B large — positively related

$\operatorname{cov} < 0$
A large when B small — negatively related

$\operatorname{cov}(A, A) = \operatorname{Var}(A)$
variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

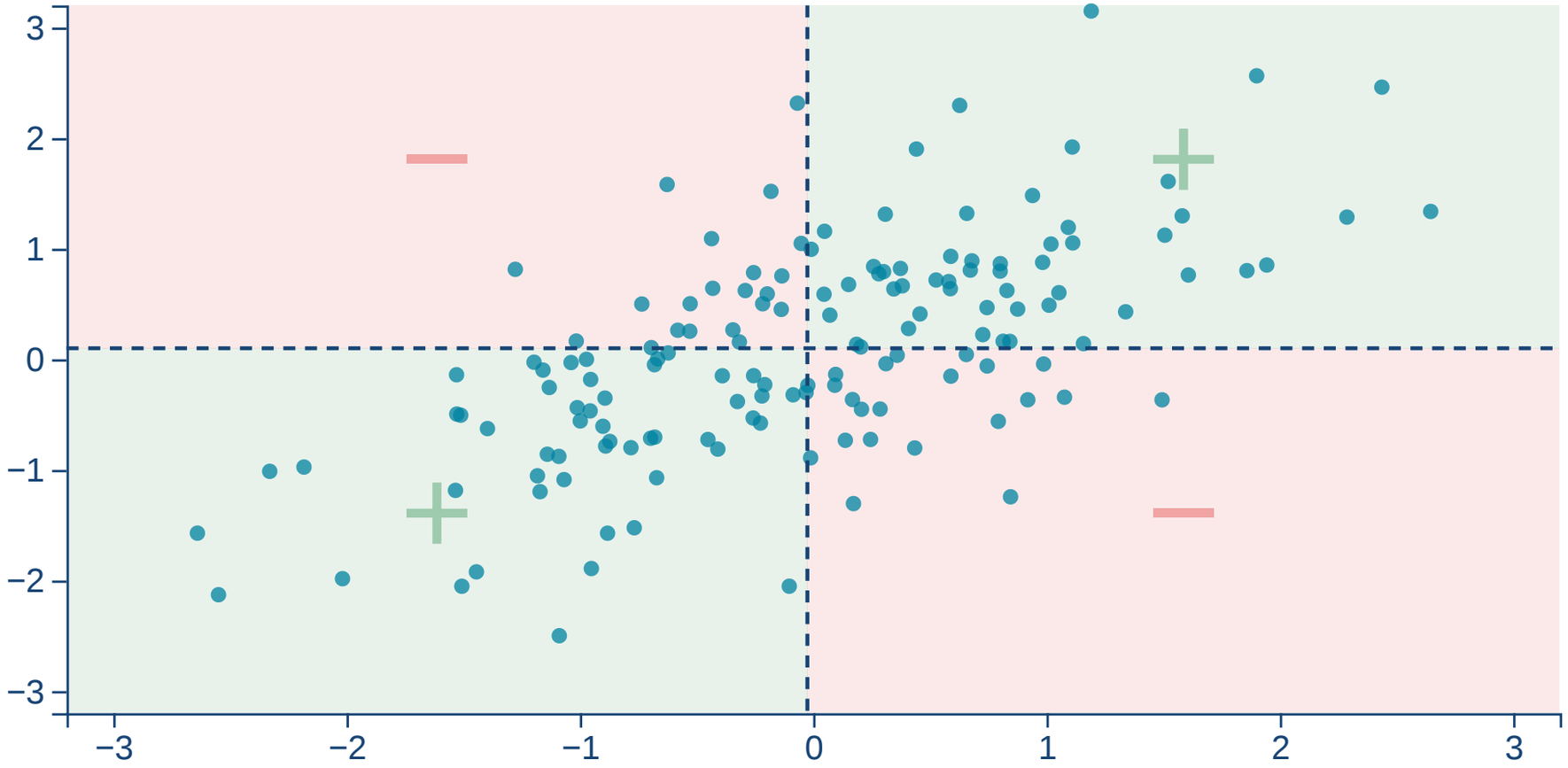
Speaker notes

The warning is the load-bearing statement. A classic counterexample: X uniform on [−1,1] and $Y = X^2$ are dependent but have zero covariance. This is why later methods that rely on correlation can miss real structure.

Covariance Sees Only One Kind of Structure

Slide ρ, then switch the relation. Predict the covariance first.

cov(X, Y) = 0.62



Predict first: with $Y = X^2 - 1$, every point is fixed by its x. The covariance?

Speaker notes

Stepped, then manipulable (P6 tiers 2 and 3). The quadrant signs make the definition visible: covariance averages the sign of (x−x□)(y−□), so a cloud that is symmetric about the vertical line has the two signs in balance and averages to zero — however tightly the points sit on a curve.

Collect the prediction before switching. Almost everyone says "large": the dependence is total. The reading afterwards is ≈ 0. That is the counterexample from the warning above, on screen, and it is the reason "uncorrelated" and "independent" must stay different words.

Takeaway

The joint distribution holds everything; **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 6

Describing Data

> ***Guiding Question:*** *The distribution is unknown and all we have is a sample. What can we honestly say about it?*

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

Population
The entire group — all humans on earth, all transactions, all documents.
Its properties (μ, σ^2) are the truth we cannot observe.

Sample
The observations we actually have.
Its properties ($\overline{x}$, s^2) are what we compute; estimates of the population's ($\overline{x}$, $\hat{\sigma}^2$) are built from them.

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

Speaker notes

This slide is the hinge of the whole module: everything before it was deduction from a model, everything after is inference from data. The population/sample distinction is exactly the gap that statistics manages.

What We Summarise

Central tendency

1st order moment
sample mean, median

Dispersion

2nd central moment
variance, standard deviation, interquartile range

Shape

3rd central moment
skewness — or, without moments, quantiles

Univariate statistics describe one random variable; **multivariate** statistics describe $X_1, \ldots, X_d$ together.

Speaker notes

The moment ordering gives a clean mental index: where is it, how spread, how lopsided. Each has a robust and a non-robust representative: the median against the mean for location, the interquartile range against the standard deviation for spread.

Means

Sample mean

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Weighted mean

$$\overline{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Geometric mean — for rates and ratios

$$\left(\prod_i x_i\right)^{1/n}$$

Harmonic mean — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

Trimmed mean: drop the k smallest and largest values, then average — a cheap way to buy robustness.

Speaker notes

The sample mean estimates E[X]. Ask when the arithmetic mean is simply wrong: average speed over equal distances (harmonic), average growth rate (geometric). Choosing the mean is a modelling decision, not a formality.

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$-th value, for even n the average of the two middle values.

Robust against outliers — one extreme value moves the median by at most one position

Costlier to compute: requires sorting, or an interval approximation over binned data

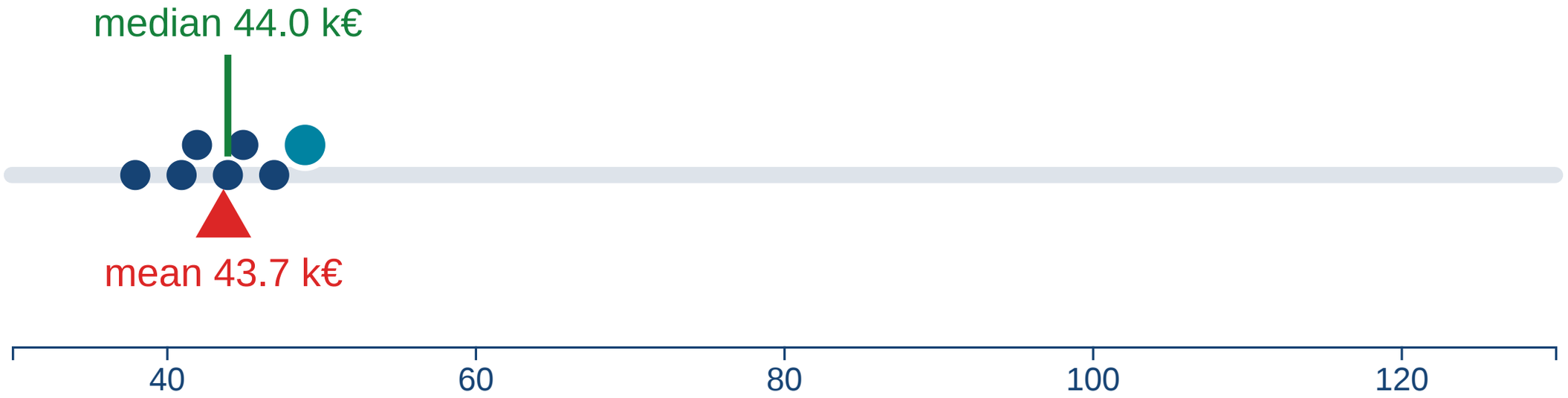
For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

Speaker notes

The mean-versus-median contrast is the first robustness discussion of the course, and it returns in clustering (k-means vs. k-medoids) and in evaluation. Ask which one a salary report should quote, and why the answer is political.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?

Speaker notes

Manipulable widget (P6 tier 3) with the prediction taken first, as the policy requires. The mean is drawn as the fulcrum the beam balances on, so the drag is visibly a lever: one value far out has to be balanced by everything else.

Drag it all the way and stop. The mean is now above six of the seven salaries — a "typical" value that almost nobody earns. The median has not moved at all — it is still 44 k€. Then ask the political question: which number goes in the recruitment advert, and which one in the union's press release?

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Speaker notes

The punchline: neither number is wrong, but only one is honest as a summary of "typical". This is the first appearance of the idea that a statistic encodes a question.

Mini-Exercise — Solution

Mean ≈ 73.9, median $= 44$.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee; the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Dispersion describes how stretched or squeezed the data is around its centre.

Sample variance — the variance *of the observed data*

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Standard deviation $s = \sqrt{s^2}$, in the unit of the data.

Computable in **one pass** over the data, $O(n)$, using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

Speaker notes

The one-pass remark is the data-engineering link: streaming aggregations rely on it. Mention the numerical-stability caveat if the group is advanced.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. If the true mean μ is unknown, we must estimate it — and using $\overline{x}$ in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Dividing by $n-1$ compensates: one degree of freedom was spent estimating the mean from the same data.

Naming is not uniform: many textbooks, and pandas, call the $n-1$ version s^2; NumPy's `var` divides by n unless told otherwise.

Speaker notes

Do not derive the correction; give the intuition. A good probe: with n = 1, the sample variance is 0 (the point equals its own mean) — clearly not an estimate of population spread, and the corrected formula refuses to answer at all.

Takeaway

s^2 describes **the sample**; $\hat{\sigma}^2$ estimates **the population**. The $n-1$ is not a convention — it is the price of having estimated the mean from the same data.

Quantiles

> The k-th q-**quantile** is the value x such that k/q of the values are smaller and $(q-k)/q$ are larger ($0 < k < q$).

Percentiles

$q = 100$

"the 12th percentile"

Quartiles

$q = 4$

Q_1 = 25th, Q_2 = 50th (= median),
Q_3 = 75th percentile

IQR

$$\text{IQR} = Q_3 - Q_1$$

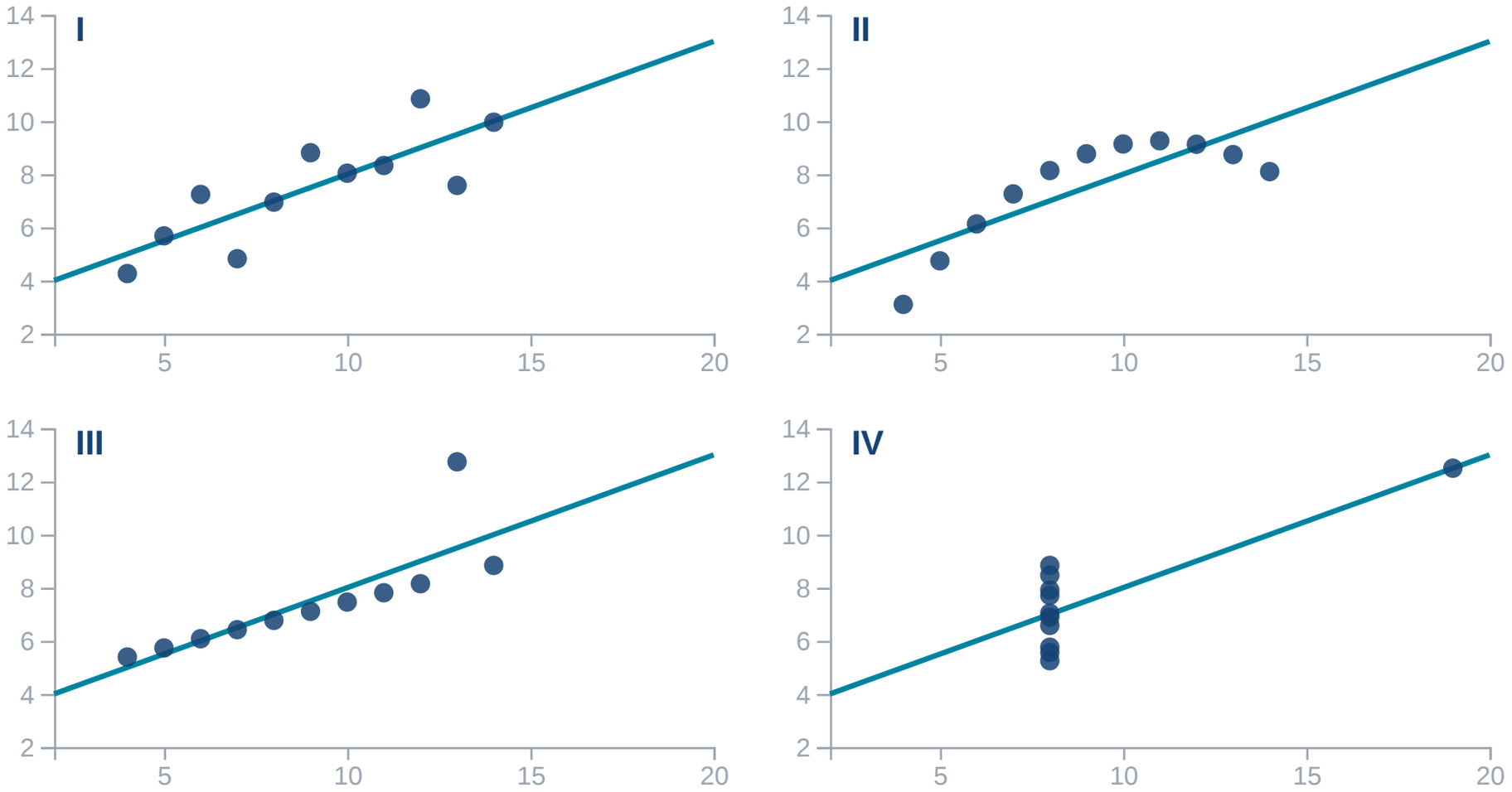
the middle half of the data

Quantiles summarise **shape and skewness** without assuming a distribution.

Speaker notes

Quantiles are the distribution-free alternative to moments: no assumption of symmetry, no sensitivity to tails. The boxplot is their standard graphical form — minimum, Q1, median, Q3, maximum, with outliers marked separately.

Anscombe's Quartet



Same mean, same variance, same correlation, same regression line — different data

all four: $\bar{x} = 9$, $s^2_x = 11$, $\bar{y} = 7.50$, $s^2_y \approx 4.12$, $r \approx 0.82$, $\hat{y} = 3.00 + 0.500 \cdot x$

Speaker notes

Stepped reveal (P6): one dataset per click, then the shared regression line and the shared statistics. All four have the same mean, variance, correlation and least-squares line. Ask after the fourth click, before revealing: "these four have identical summary statistics — what does that tell you about summary statistics?" Anscombe (1973), the origin of "always plot your data".

What the Quartet Teaches

Summaries compress — and compression loses.
Identical moments can hide a curve, an outlier, or a variable that barely varies at all.

Plot first, summarise second.
Visual inspection is not a soft alternative to statistics; it is how you find out which statistics are meaningful.

Quantiles describe a distribution without assuming its shape — and no set of summary numbers can replace looking at the data.

Speaker notes

This is the bridge to the data-exploration chapter, where visual analytics gets its own treatment. Stabilise the lesson as a working rule, not a slogan: every summary answers one question and is silent about the rest.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

Speaker notes

Open on the gap, not on a definition (P1, P4). Most students will answer with a number or hedge; the point is that the question is malformed. Everything in this unit exists to replace "μ = 73.9" with "μ = 73.9 ± something", and to say where the something comes from.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

Zero. $\overline{x}$ is not μ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is* μ but **how far from** μ **can this number be?**

Estimator and Estimate

Definition. An **estimator** is a function of the sample — $\hat{\theta} = g(X_1, \ldots, X_n)$ — used to approximate an unknown quantity θ of the population. The number it takes on for one particular sample is an **estimate**.

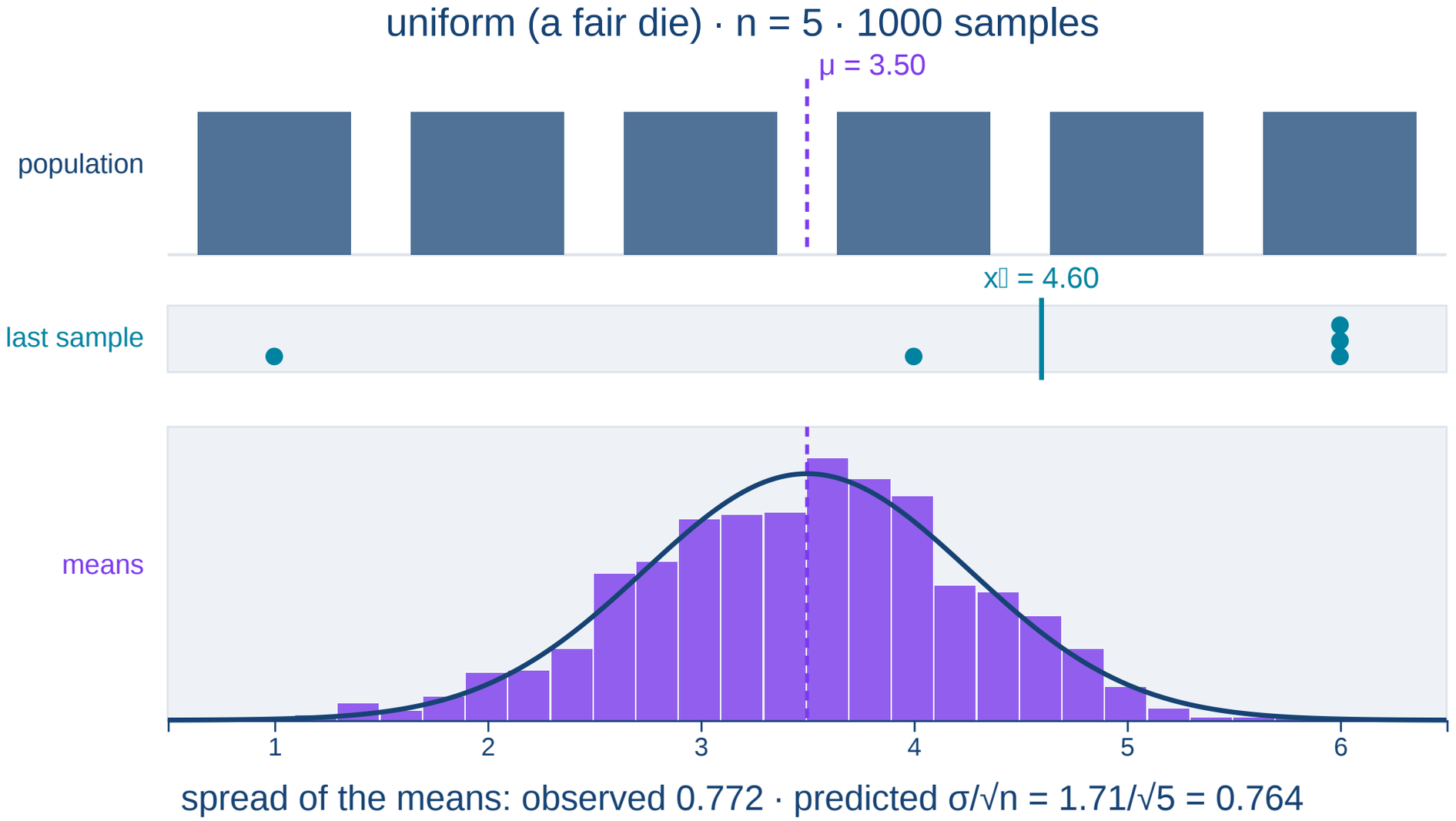
The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

$\overline{x}$, s^2 and $\hat{\sigma}^2$ from descriptive statistics are all estimators. We used them without saying what kind of object they are.

Speaker notes

Math first (P5) — this is a second pass over objects the students already compute. The load-bearing sentence is the middle one: a statistic inherits randomness from the sample. Everything else follows from taking that seriously.

An Estimator Has a Distribution



Speaker notes

Stepped visualization (P6 tier 2 with tier-3 controls). One click per step: draw one sample; record its mean; fifty samples; a thousand, with the normal curve and the standard error. Both pictures share the x-axis, so "the means are more tightly packed than the population" is something students see rather than something we assert. Collect predictions before the third click — the common guess is that the spread stays the same.

Three Things the Picture Shows

- → The means pile up **around** μ — the estimator is not systematically off
- → The pile is **narrower than the population** — averaging cancels noise
- → The pile is **bell-shaped**, even though the population was not

> **Definition (Sampling distribution).** The distribution of an estimator $\hat{\theta}$ over all possible samples of size n. $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Speaker notes

The three observations are, in order: unbiasedness, the standard error, and the central limit theorem — named on the next slides. Naming them here would be faster and worse. Ask which of the three would still hold for the *median* (the first two do; the third does too, but that needs a different argument).

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

Speaker notes

Tier-3 exploration, with the prediction committed first as the policy requires. The bimodal population is the persuasive one: two separate humps, and the means of five draws still form a single bell. Push n to 2 to show where it breaks down — the claim is asymptotic, not magic.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as n is more than a handful — which is why one formula can cover cases we never inspect.

Standard Error

Definition (Standard error). The standard deviation of an estimator's sampling distribution. For the sample mean of independent draws:

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

It is measured in the unit of the data and answers "how far off is a typical estimate", where σ answers "how far apart are typical *observations*".

σ and SE are different quantities. More data does nothing to σ — the world stays as variable as it was — and shrinks SE.

Speaker notes

The warning is the most valuable line in the unit: σ is the variability of the world (aleatoric uncertainty), SE is our ignorance about the estimate (epistemic uncertainty) — the two words return at the end of the unit. Ask which one a weather forecast's "±3 °C" refers to; the answer is usually both, mixed.

Precision Costs Data

Multiply the data by…	…and the standard error is multiplied by
2	0.71
4	**½**
25	⅕
100	**1⁄10**

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Speaker notes

The economics of data collection in one line, and a sanity check students can apply to any claim about "more data". Ask what this implies for an A/B test that wants to detect a 0.1% difference. It also foreshadows why we later care about better *models* rather than only bigger *samples*.

Two Theorems We Have Already Seen Happen

Law of large numbers

$\overline{X}_n \to \mu$ as $n \to \infty$.

The estimate converges on the truth — the pile keeps narrowing.

Central limit theorem

$\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$ for large n.

And we know the *shape* of what is left.

The first says the error goes away; the second says how much of it is left after n observations. Only the second lets us put a number on our ignorance.

Speaker notes

Deliberate spiral (P7): the CLT was stated in the continuous-distributions unit and was visible in the binomial sweep. Here it acquires a job. Emphasise the division of labour between the two theorems — students routinely quote the LLN where they need the CLT.

From a Number to an Interval

If $\overline{X} \approx N(\mu, \sigma^2/n)$, then for 95% of samples $\overline{X}$ lands within $1.96\,\mathrm{SE}$ of μ — and therefore μ lands within $1.96\,\mathrm{SE}$ of $\overline{X}$.

$$\overline{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

the estimate — our best single guess

how confident we insist on being

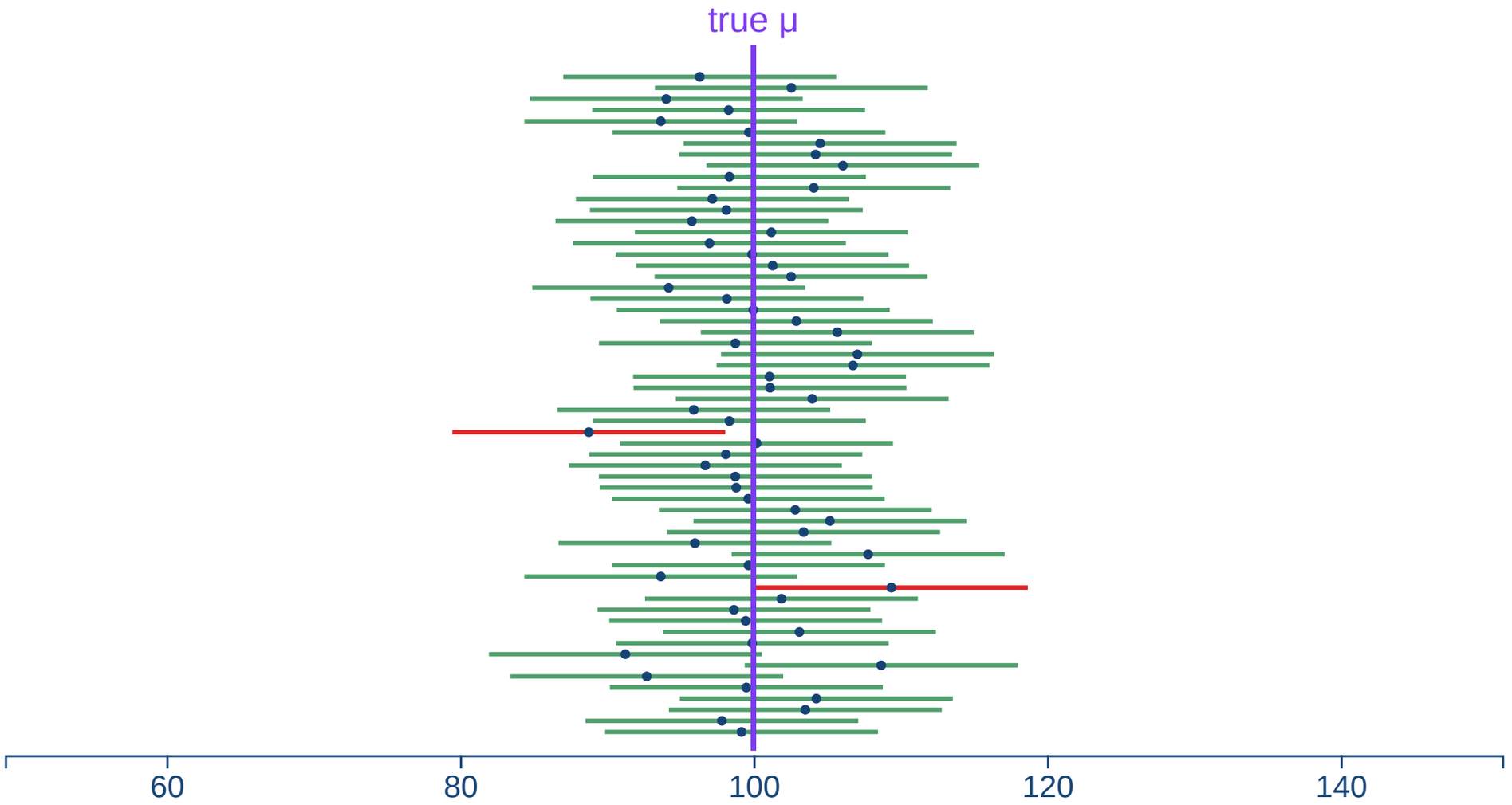
how noisy the estimate is

Speaker notes

Math-first-then-meaning (P5), legitimate because the picture came first. Walk the three parts and ask what each one would have to do to make the interval wider. σ is treated as known here; the correction for an estimated σ (the t distribution) is a refinement, not a different idea.

What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?



58 of 60 intervals cover μ (97%) — nominal 95%

Speaker notes

Stepped, then manipulable. One interval; twenty; sixty with the coverage tally; then the invitation to change the level and n. Ask before the third click how many of sixty intervals should miss. The red rows are the whole argument: the misses look exactly like the hits from the inside.

The Sentence Almost Everyone Gets Wrong

WHAT IT DOES NOT MEAN

"There is a 95% probability that μ lies in **this** interval."

μ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain μ."

The probability is a property of the recipe, evaluated before the data is seen.

Speaker notes

This is the distinction the picture just made — the rows are the procedure, one row is your data. Expect resistance: the wrong reading is the useful-sounding one. Note also that a Bayesian credible interval *does* license the first sentence, at the price of requiring a prior — a forward pointer, not a dependency.

Where This Leaves Us

Two kinds of uncertainty meet in one formula. We can now measure one of them.

- → σ — how variable the world is. **Aleatoric.** Unchanged by data.
- → $\mathrm{SE} = \sigma/\sqrt{n}$ — how badly we know our own estimate. **Epistemic.** Shrinks with data.

"More data helps" was a slogan.
It is now a rate: $1/\sqrt{n}$.

Speaker notes

If the course has already named the two kinds of uncertainty, close that loop here; if not, the two bullets introduce the words. This is the slide to point back to whenever a later lecture claims a model is uncertain "because it has not seen enough" — that claim has a magnitude, and this is where it comes from.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

Speaker notes

Solution on the same slide, click-gated. The second part is the one that transfers: the √n law is the answer to nearly every "how much data do we need" question a student will be asked. The interval is 180 ± 1.96 · 6 ≈ [168, 192] s. If time allows, ask what would change if σ had to be estimated too.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

$\mathrm{SE} = 60/\sqrt{100} = 6$ s, so the 95% interval is about 180 ± 12 s. Halving the SE needs twice the $\sqrt{n}$ — **four times the data**, $n = 400$. $\sigma = 60$ s does not change: customers stay as variable as they were; only our knowledge of their average improves.

Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.

Speaker notes

Stabilise three things: statistic-as-random-variable, the 1/√n rate, and the procedure reading of confidence. All three return whenever a measured number — an accuracy, an error rate — is reported with an error bar, and a Bayesian treatment gives the same question a different, more directly stated answer.

Module 7

Hypothesis Testing

> ***Guiding Question:*** *Your new model scores higher than the baseline. How do you know the improvement is real?*

Motivation: Comparing Two ML Experiments

Experiment A — baseline model accuracy measured over n_A runs

Experiment B — new architecture accuracy measured over n_B runs

B scored higher. Is the improvement systematic, or just randomness?

H_0: the two means are equal — $\mu_A = \mu_B$
H_1: they differ — $\mu_A \neq \mu_B$

Speaker notes

Guiding question of the module, in the students' own working context. Every one of them has compared two model runs; almost none has asked whether the difference survives resampling. Keep the example running through the unit.

The Idea

> Compare what we **observed** with what H_0 **predicts**. If the disagreement is large enough to be implausible under H_0, reject H_0.

We never prove H_1. We only measure how badly H_0 fails to explain the data.

Speaker notes

The asymmetry is the hardest idea in the unit. "Fail to reject" is not "accept". Ask what a court verdict of "not guilty" claims, and what it does not.

Test Statistic and p-Value

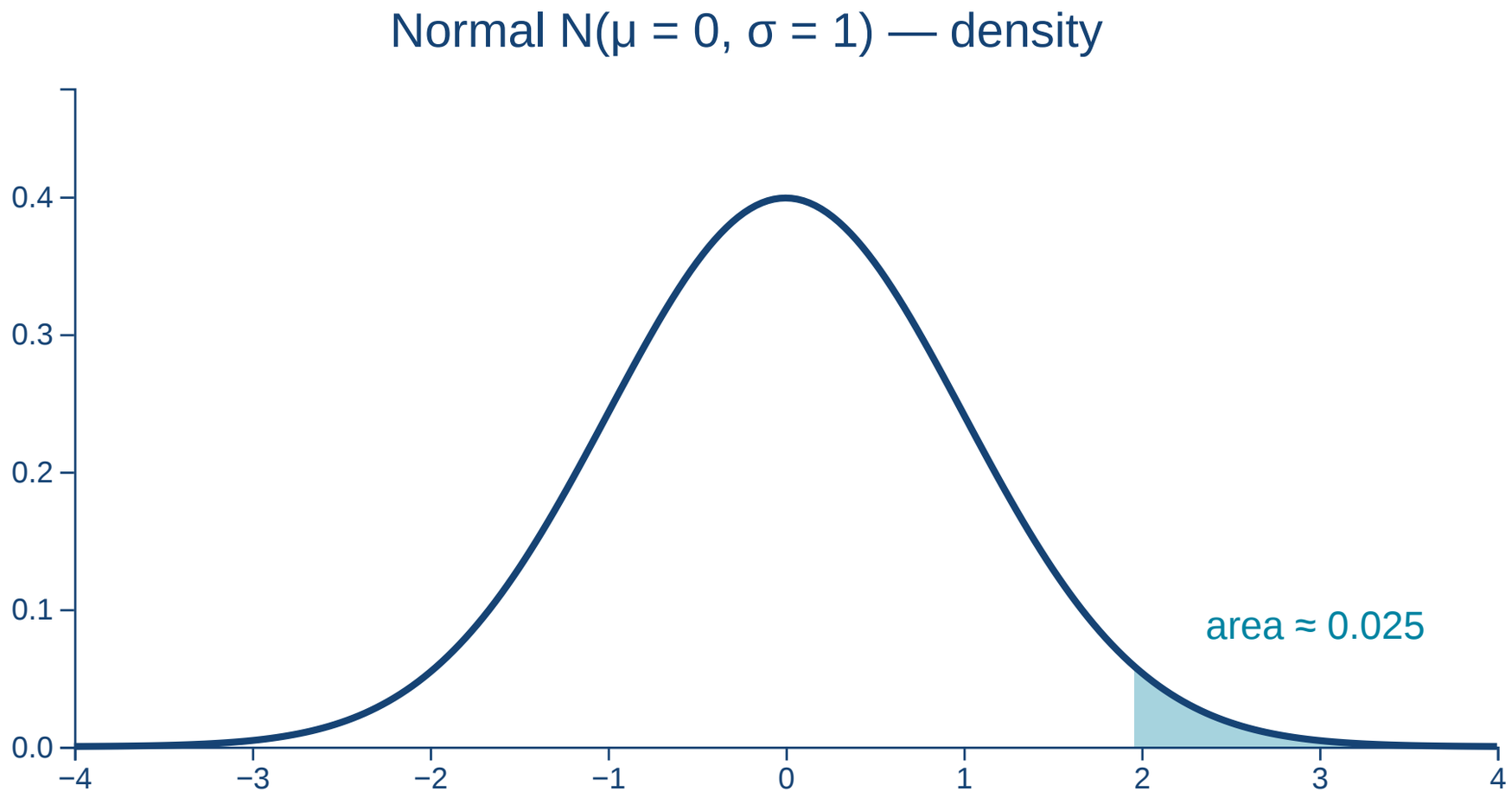
A **test statistic** $T = T(X_1, \ldots, X_n)$ compresses the data so that its distribution **under** H_0 is known.
The **p-value** is the probability, under H_0, of observing a statistic at least as extreme as the one we got.
Small p-value → the data is surprising under H_0.

> The p-value is **not** the probability that H_0 is true. It is $P(|T| \geq |t_{\text{obs}}| \mid H_0)$ — a tail probability of the statistic — not $P(H_0 \mid \text{data})$.

Speaker notes

The warning box is the Bayes unit returning: swapping the two conditionals is the same error as the disease test, now with a career's worth of published consequences. Make that link explicit — students remember it.

The Rejection Region



Under H_0 the statistic follows a known distribution. The shaded tail carries probability $\alpha/2$: if T lands there, we reject.

Speaker notes

One click shades the tail. Emphasise: α is chosen *before* seeing the data, and it is the rate of false alarms we are willing to tolerate — a decision, not a discovery.

Two Ways to Be Wrong

Type I error — false alarm
Reject H_0 when it is true.
Probability α (we choose it).

Type II error — miss
Fail to reject H_0 when H_1 holds.
Probability β; **power** $= 1 - \beta$.

Lowering α raises β for a fixed design. To improve both: more data, less noise, or a **paired** design that cancels shared noise.

Speaker notes

The trade-off is the practical content of the slide. In an ML context: a strict α means real improvements get dismissed; a lax α means noise gets published. Power analysis answers "how many runs do I need", which is the question people actually have. The paired design is the cheap one in ML: evaluate both models on the same seeds and test splits, and the split-to-split noise cancels in the differences.

One Sample: z or t?

Known variance — z-test

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$$

Reject (two-sided) if $|Z| > z_{1-\alpha/2}$.

Unknown variance — t-test

$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$

s estimated from the data → **heavier tails** (t with $n-1$ d.f.).

$\sigma/\sqrt{n}$ is the **standard error**: the standard deviation of the sample mean $\overline{X}$. $s = \sqrt{s^2}$ is the sample standard deviation. Both statistics measure the distance from μ_0 in standard errors.

Speaker notes

The only difference between the two is the denominator. Plugging in an estimate s for σ adds uncertainty, which the t distribution's heavier tails pay for; with $n \geq 30$ the two are practically the same.

Two Samples: Which t?

Situation	Test	Statistic
independent runs, equal variances credible	pooled t	$\frac{\overline{X}_B - \overline{X}_A}{s_p\sqrt{1/n_A + 1/n_B}}$
independent runs, variances may differ	**Welch's t**	$\frac{\overline{X}_B - \overline{X}_A}{\sqrt{s_A^2/n_A + s_B^2/n_B}}$
both models on the **same** seeds or splits	**paired t**	t-test on the differences $d_i = x_{B,i} - x_{A,i}$

> Welch is the default for independent runs. When both models can be run on the same splits, pair them: the shared noise cancels in d_i.

Speaker notes

A table because the three tests differ along the same dimensions. Welch by default because the equal-variance assumption is rarely justified across architectures. The paired test is the one ML comparisons usually need and rarely use: a cross-validation comparison evaluates both models on the same folds, so the fold results are not independent samples.

Testing Variances

Single variance (normal model), χ^2 statistic with $n-1$ degrees of freedom.
Is run-to-run variability within specification?

Two variances, $F = s_A^2 / s_B^2$.
If the variances differ, **do not pool —** go back to Welch's t for the means.

Speaker notes

Variance tests matter in ML more than people think: a model that is equally accurate on average but wildly less stable across seeds is a different product.

Back to the A/B Question — Worked

Ten runs each. A: mean accuracy 80 %, $s_A = 2$. B: mean 82 %, $s_B = 1$.

$$t = \frac{82 - 80}{\sqrt{4/10 + 1/10}} = \frac{2}{0.71} \approx 2.83$$

Speaker notes

The numbers are chosen so the standard error is $\sqrt{0.5} \approx 0.71$. Welch's d.f.: $0.25/((0.16 + 0.01)/9) \approx 13.2$. The paired numbers are hypothetical — the point is that the per-seed differences vary far less than either model's accuracies do, because a hard seed is hard for both.

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Welch's degrees of freedom ≈ 13; the critical value $t_{0.975} \approx 2.16$. Since $2.83 > 2.16$, **reject** H_0 at $\alpha = 0.05$ ($p \approx 0.014$).

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> Paired on the same ten seeds, with differences of mean 2 and $s_d = 0.5$: $t = 2/(0.5/\sqrt{10}) \approx 12.6$. Pairing buys far more power than more runs.

The Recommendation

Default

Welch's two-sample t on the mean difference — or paired t if both ran on the same splits

Care about stability

F-test on s_A^2 / s_B^2

Normality dubious

Permutation (randomisation) test on the mean difference

Hypothesis testing compares data to a model through a statistic whose null distribution is known. Known vs. unknown variance changes that distribution (z vs. t); the design changes the test (Welch for independent runs, paired for shared splits).

Speaker notes

Close by returning to the opening question with a concrete recommendation — the module started with a decision they have to make, and ends with the decision rule. Mention that the permutation test needs no distributional assumption at all, which is why it is a good fallback when in doubt.

The Churn Shop: Is the Difference Real?

Lecture 01's dashboard, with counts. Over one year:

	customers	churned	rate
app users	400	40	10%
no app	400	60	15%

H_0: both groups churn at the same rate. Pooled rate $\hat{p} = 100/800 = 0.125$.

Speaker notes

The running example of the course (a subscription business, introduced in the Lecture 01 reasoning-cycle exercise), now with numbers. The statistic is the z-test of this lecture applied to two proportions: a proportion is the mean of a Bernoulli variable, so its standard error is $\sqrt{(p(1-p)/n)}$, and under H_0 both groups share one p, estimated from all 800 customers. Exact values: SE ≈ 0.0234, z ≈ 2.14, two-sided p ≈ 0.033.

Say the limit clearly: the test says the gap is unlikely under "no difference"; it says nothing about which way the arrow points. Loyal customers may simply be the ones who install the app.

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$$z = \frac{0.15 - 0.10}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{400} + \frac{1}{400}\right)}} = \frac{0.05}{0.023} \approx 2.1 \qquad p \approx 0.03$$

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> Chance alone would rarely produce a gap this large: at $\alpha = 0.05$ the difference is **real**. Whether the app *causes* it is a different question — Lecture 03 draws both answers.

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.